



POLITECNICO
MILANO 1863

Robotics

Robot Localization – Wheels Odometry

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Where Am I?

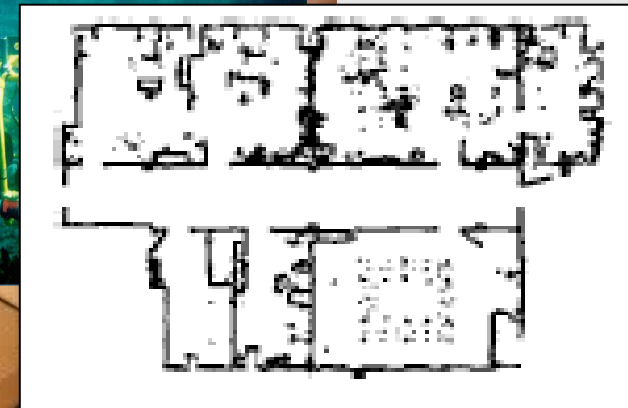
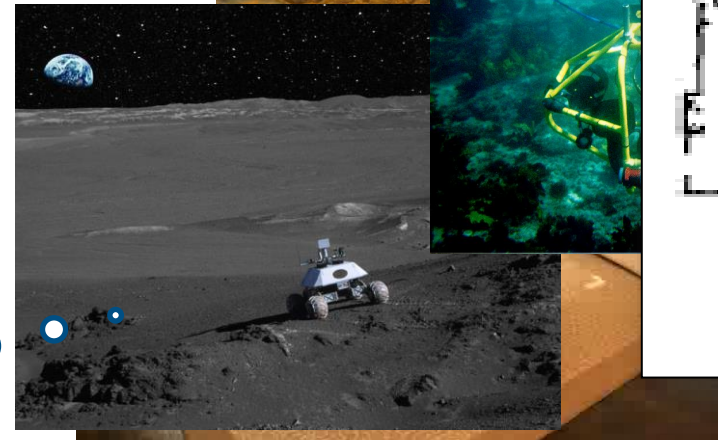
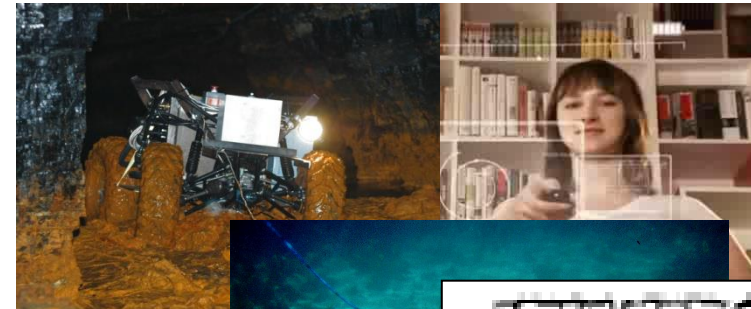
To perform their tasks autonomous robots and unmanned vehicles need

- To know where they are (e.g., Global Positioning System)
- To know the environment map (e.g., Geographical Institutes Maps)

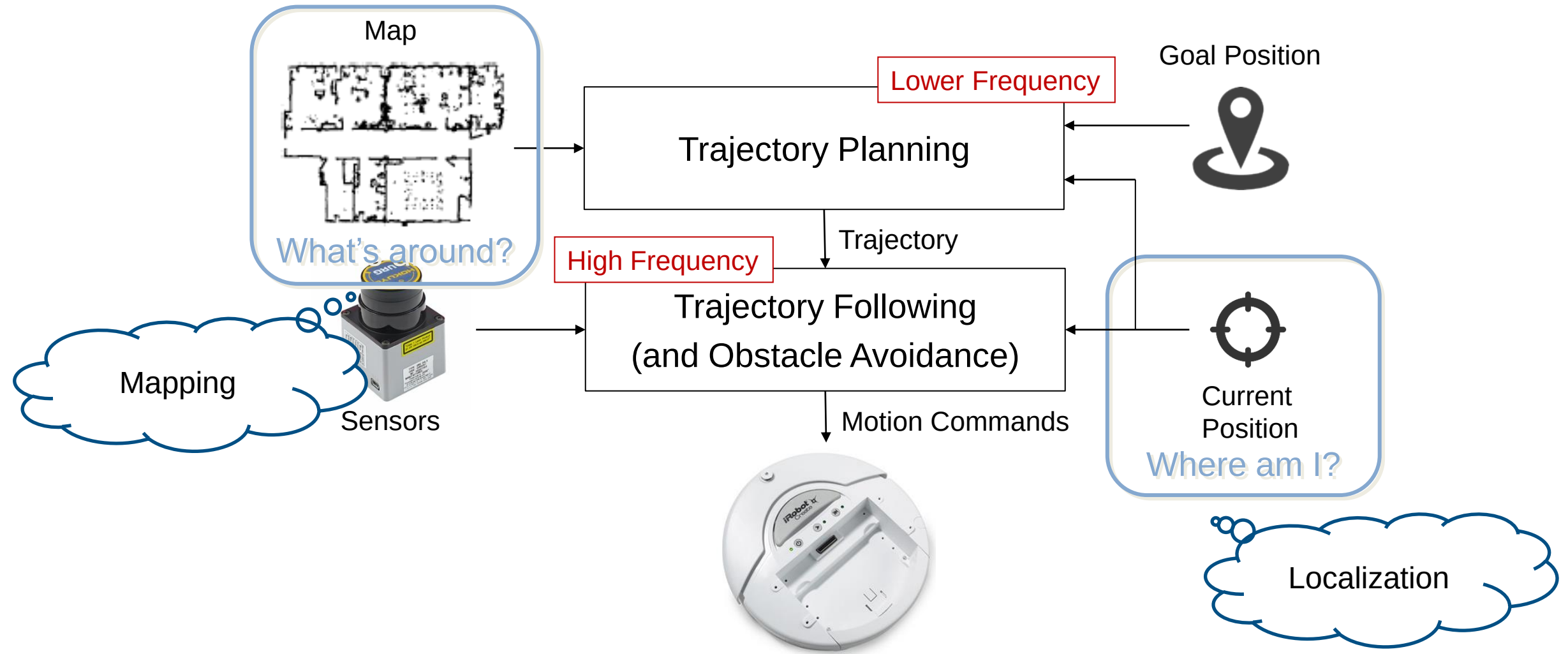
These are not always possible or reliable

- GNSS are not always reliable/available
- Not all places have been mapped
- Environment changes dynamically
- Maps need to be updated

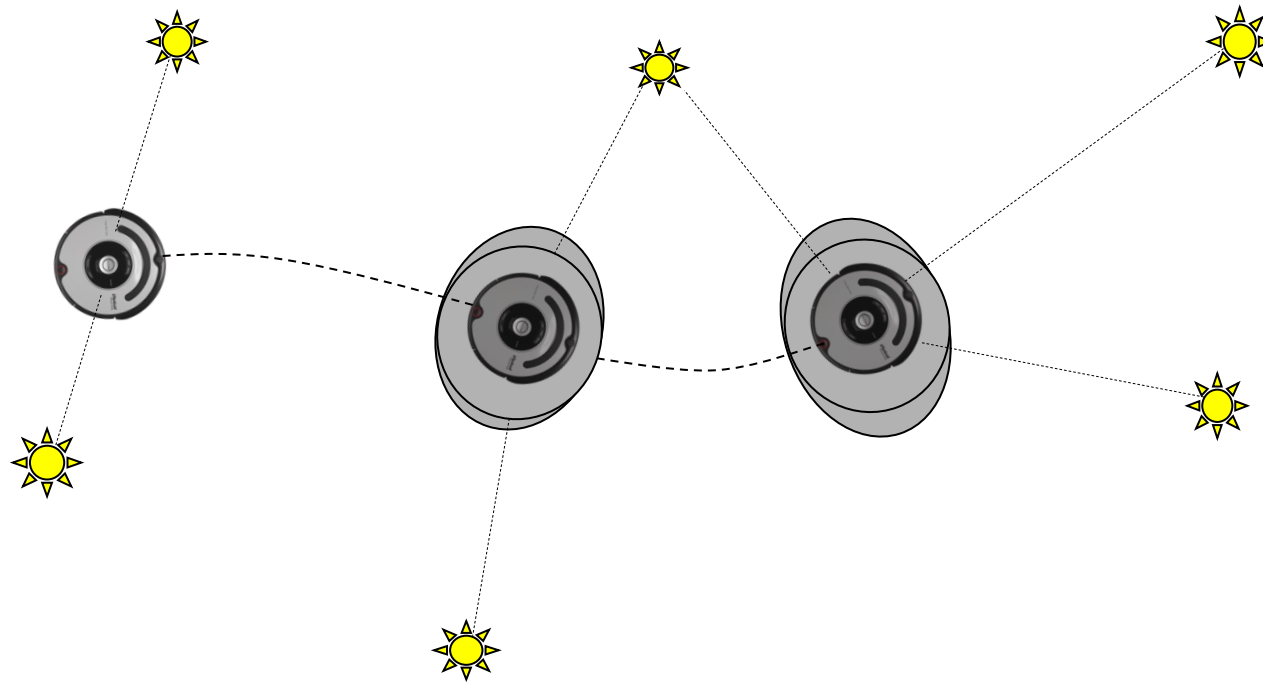
How would you
program this robot?



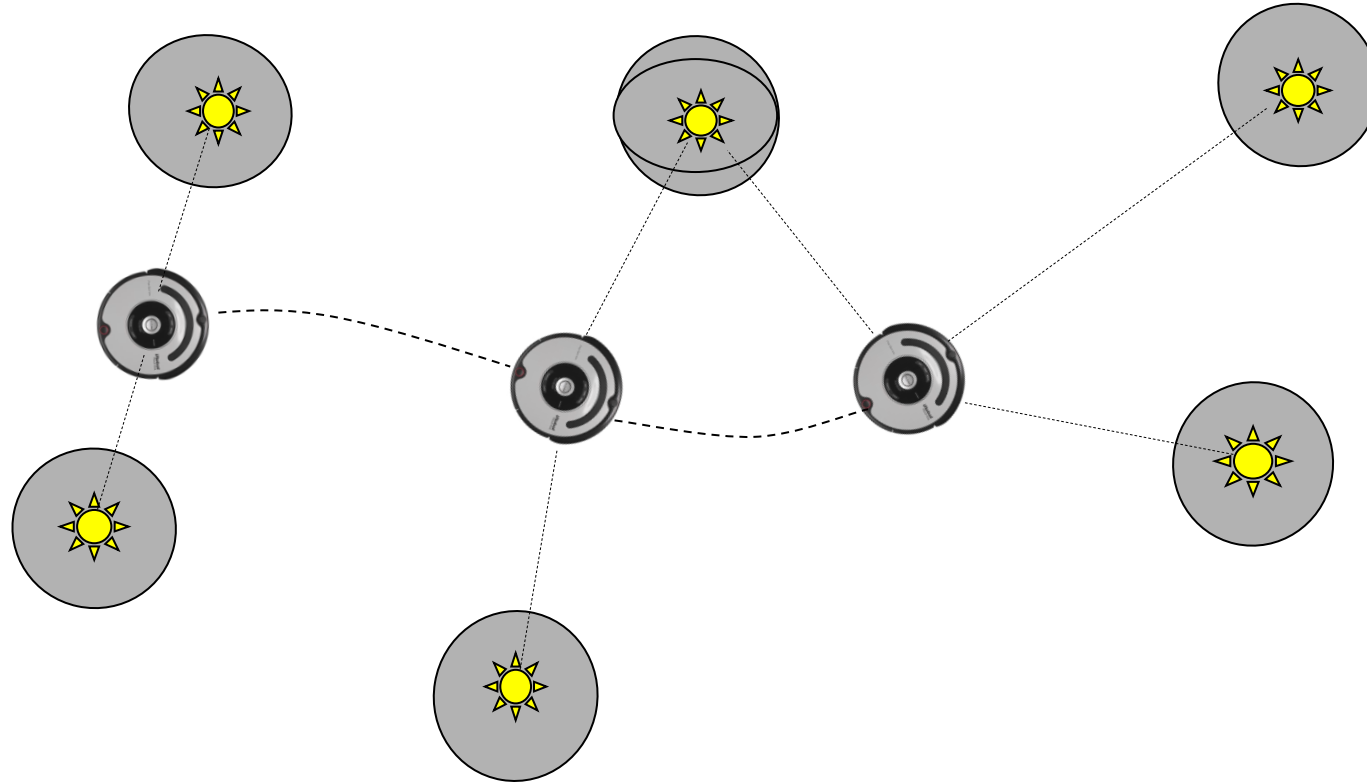
A Simplified Sense-Plan-Act Architecture



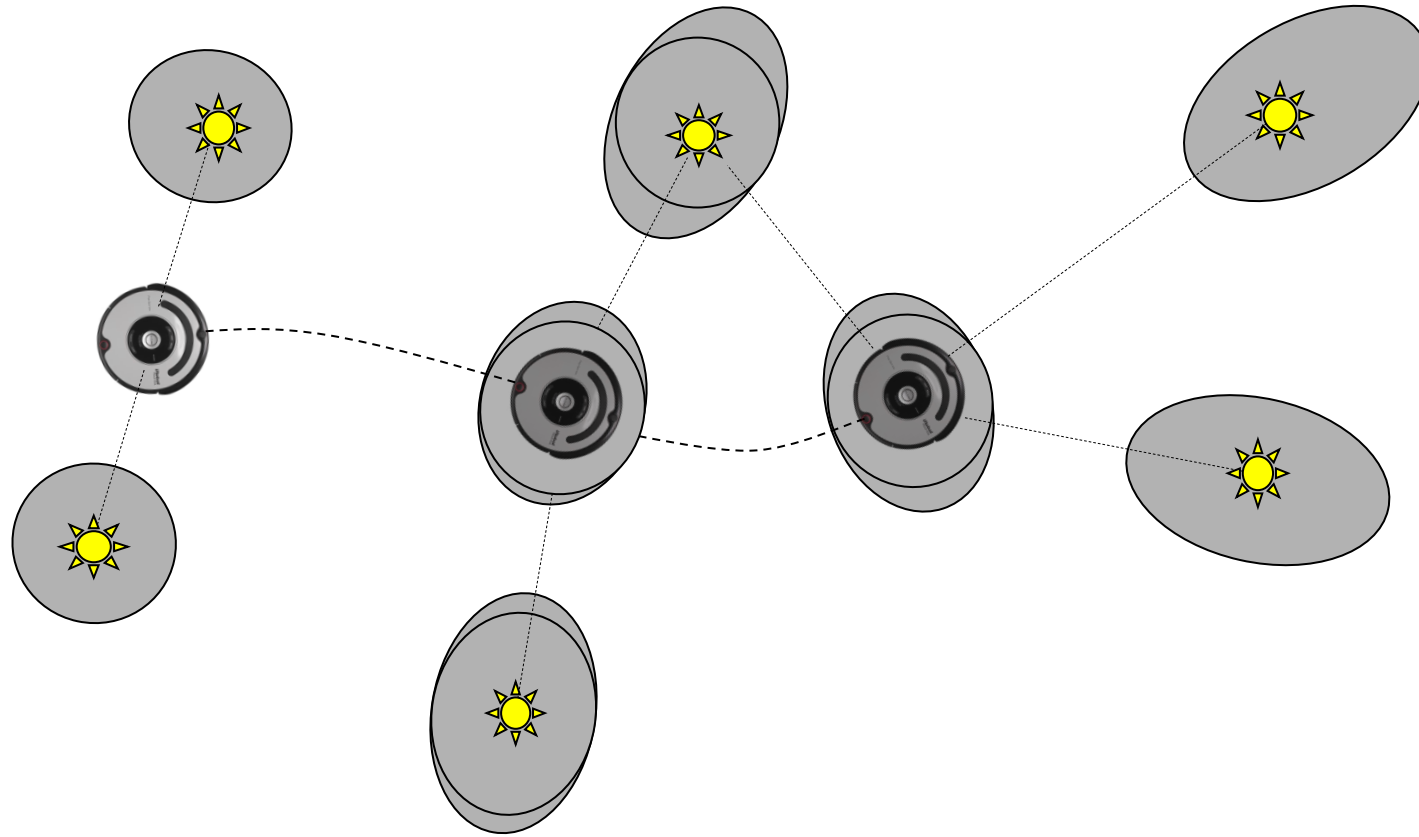
Localization with Known Map



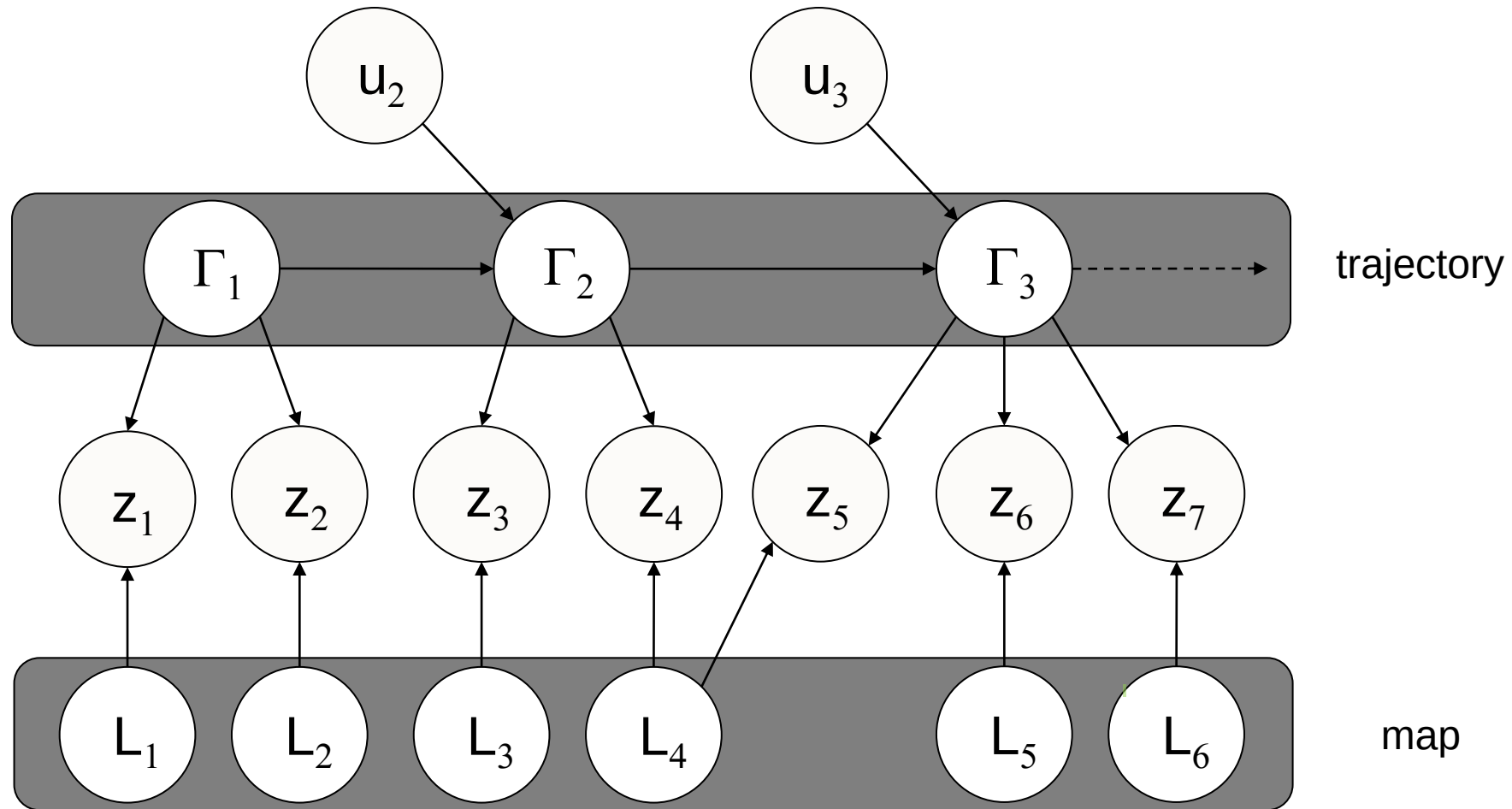
Mapping with Known Poses



Simultaneous Localization and Mapping

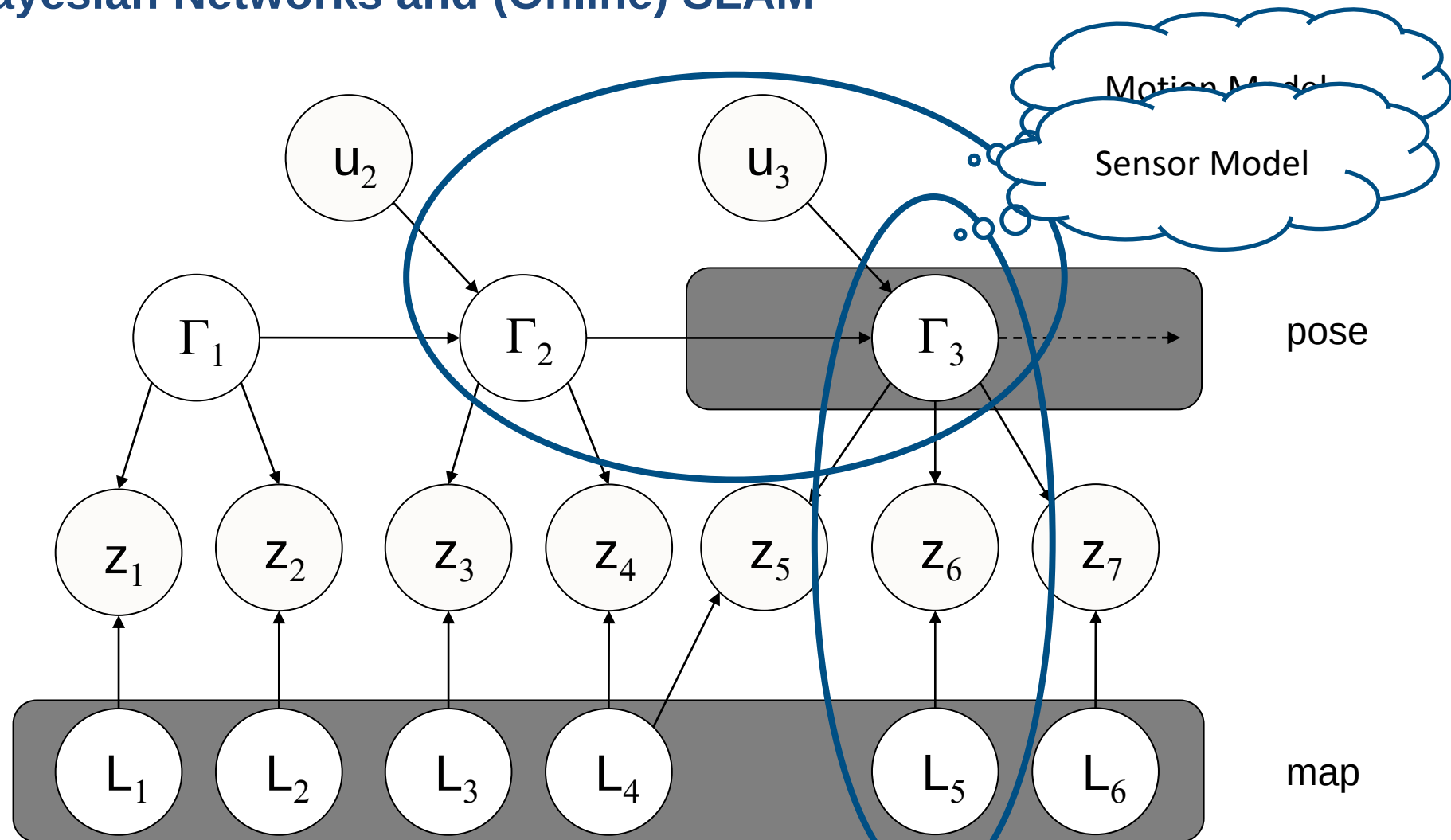


Dynamic Bayesian Networks and (Full) SLAM



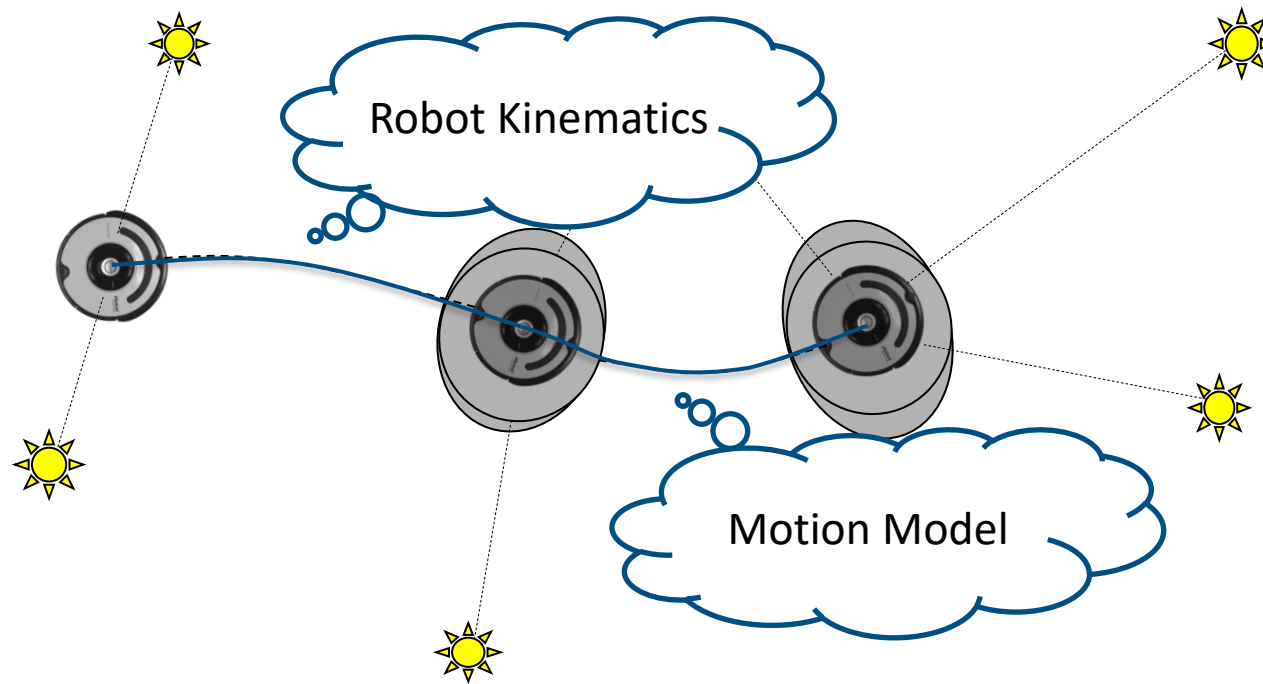
$$\text{Smoothing : } p(\Gamma_{1:t}, l_1, \dots, l_N \mid Z_{1:t}, U_{1:t})$$

Dynamic Bayesian Networks and (Online) SLAM



Filtering :
$$p(\Gamma_t, l_1, \dots, l_N | Z_{1:t}, U_{1:t}) = \iiint_{1:t-1} p(\Gamma_{1:t}, l_1, \dots, l_N | Z_{1:t}, U_{1:t})$$

Localization with Known Map



Wheeled Mobile Robots

A robot capable of locomotion on a surface **solely through the actuation of wheel assemblies** mounted on the robot and in contact with the surface. A wheel assembly is a device which provides or allows motion between its mount and surface on which it is intended to have **a single point of rolling contact.**

(Muir and Newman, 1986)



Robot Mobile



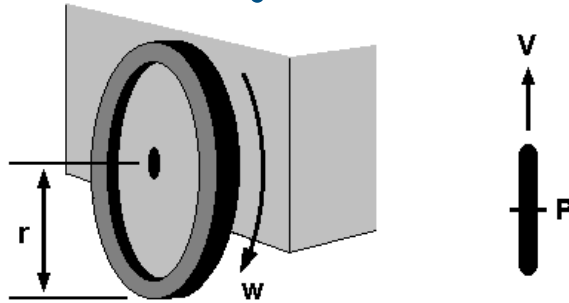
AGV



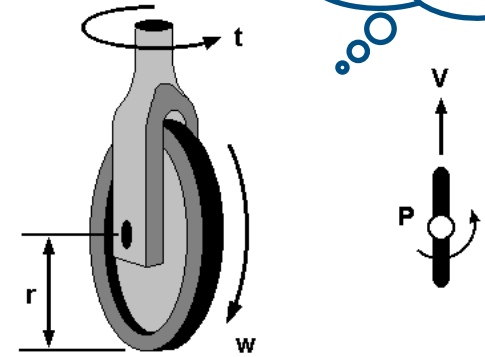
Unmanned vehicle

Wheels Types

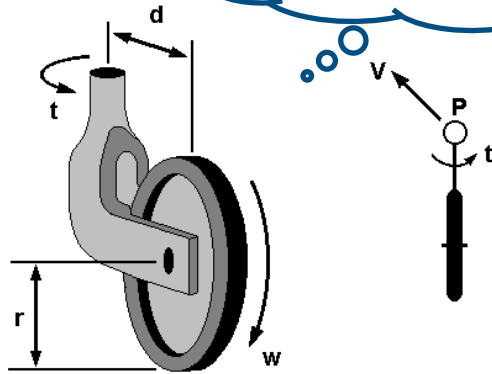
Fixed



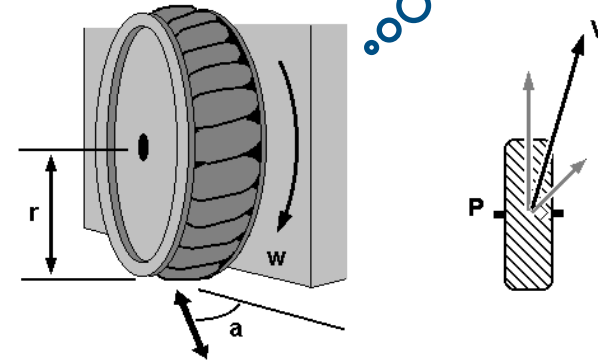
Orientable centered



Caster omnidirectional



Swedish or Meccanum



Mobile Robots Types/Kinematics (some)

Two wheels (differential drive)

- Simple model
- Suffers terrain irregularities
- Cannot translate laterally



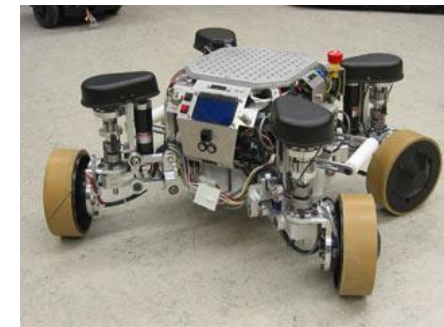
Tracks

- Suited for outdoor terrains
- Not accurate movements (with rotations)
- Complex model
- Cannot translate laterally



Omnidirectional (synchro drive)

- Can exploit all degrees of freedom (3DoF)
- Complex model
- Complex structure



Differential Drive (MRT – Politecnico di Milano)



Differential Drive (MRT – Politecnico di Milano)



Omnidirectional (Swedish wheels)



Omnidirectional (Syncro drive)



Some Definitions ...

Locomotion: the process of causing an autonomous robot to move

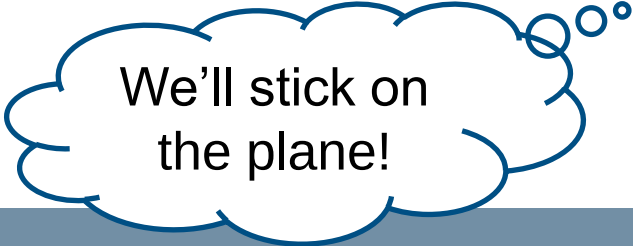
- To produce motion, forces must be applied to the vehicle

Dynamics: the study of motion in which forces are modeled

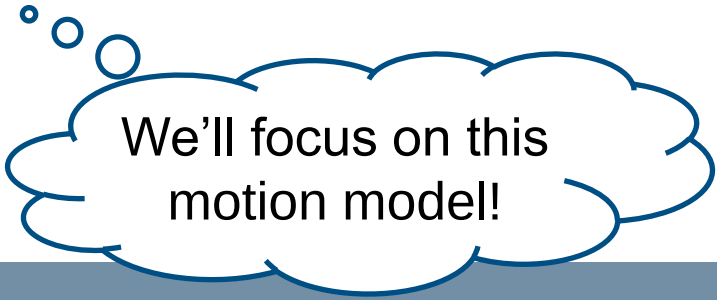
- Includes the energies and speeds associated with these motions

Kinematics: study of motion without considering forces that affect it

- Deals with the geometric relationships that govern the system
- Deals with the relationship between control parameters and the behavior of a system in state space

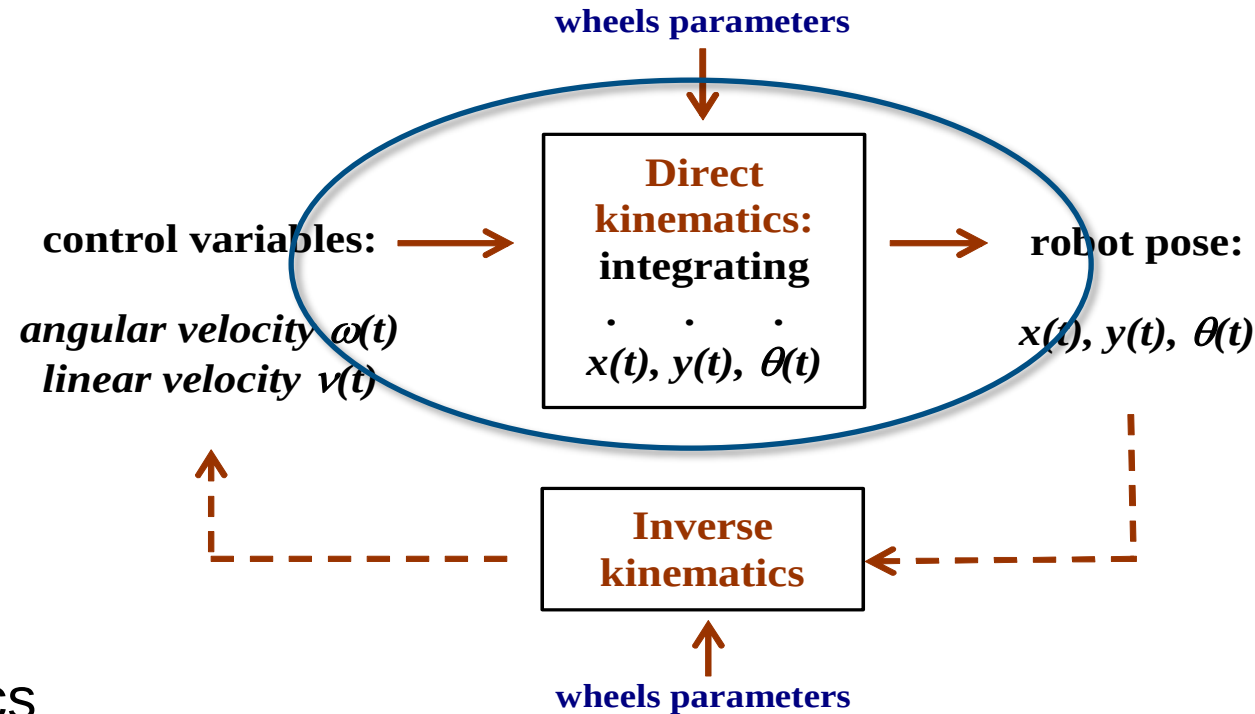


We'll stick on
the plane!



We'll focus on this
motion model!





Direct kinematics

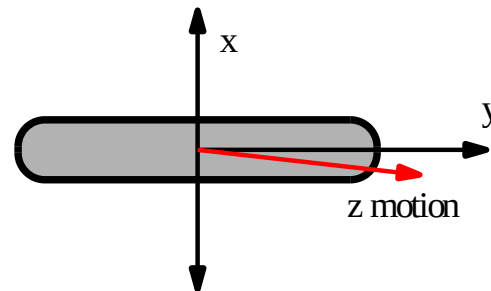
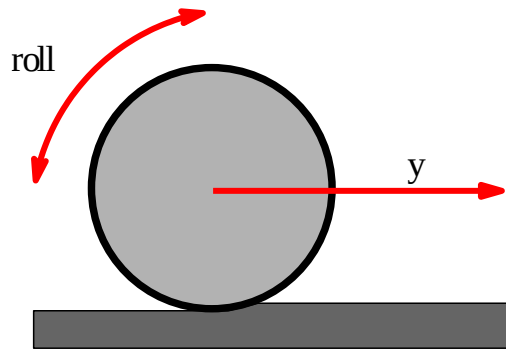
- Given control parameters, e.g., wheels and velocities, and a time of movement t , find the pose (x, y, θ) reached by the robot

Inverse kinematics

- Given the final pose (x, y, θ) find control parameters to move there in a given time t

Wheeled robot assumptions

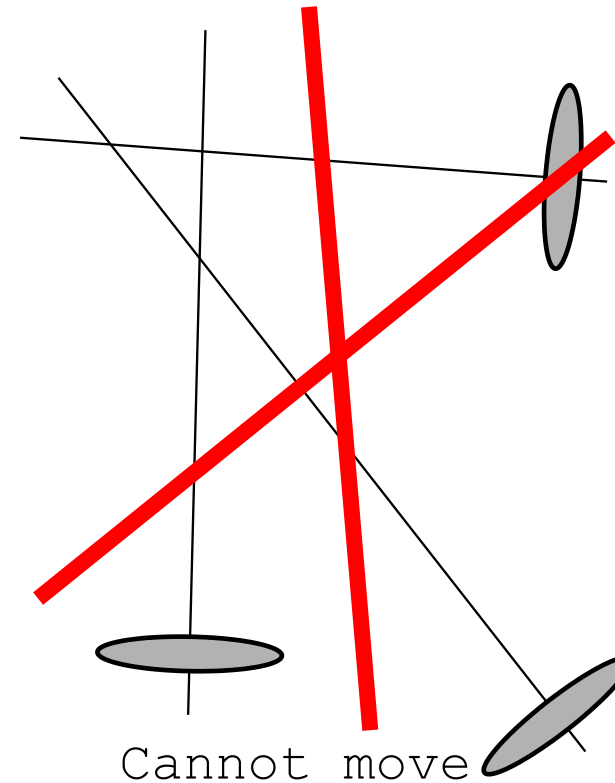
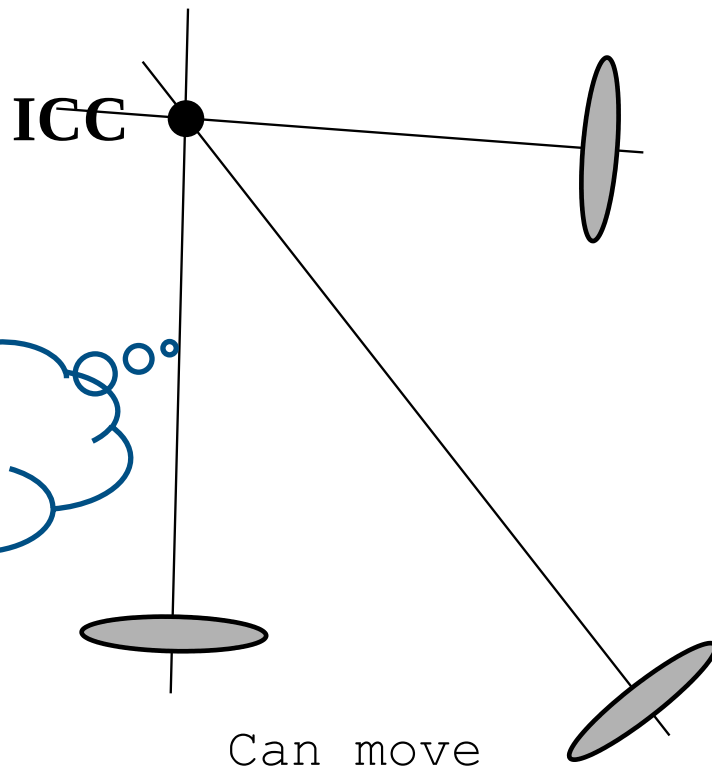
1. Robot made only by rigid parts
2. Each wheel may have a 1 link for steering
3. Steering axes are orthogonal to soil
4. Pure rolling of the wheel about its axis (x axis)
5. No translation of the wheel



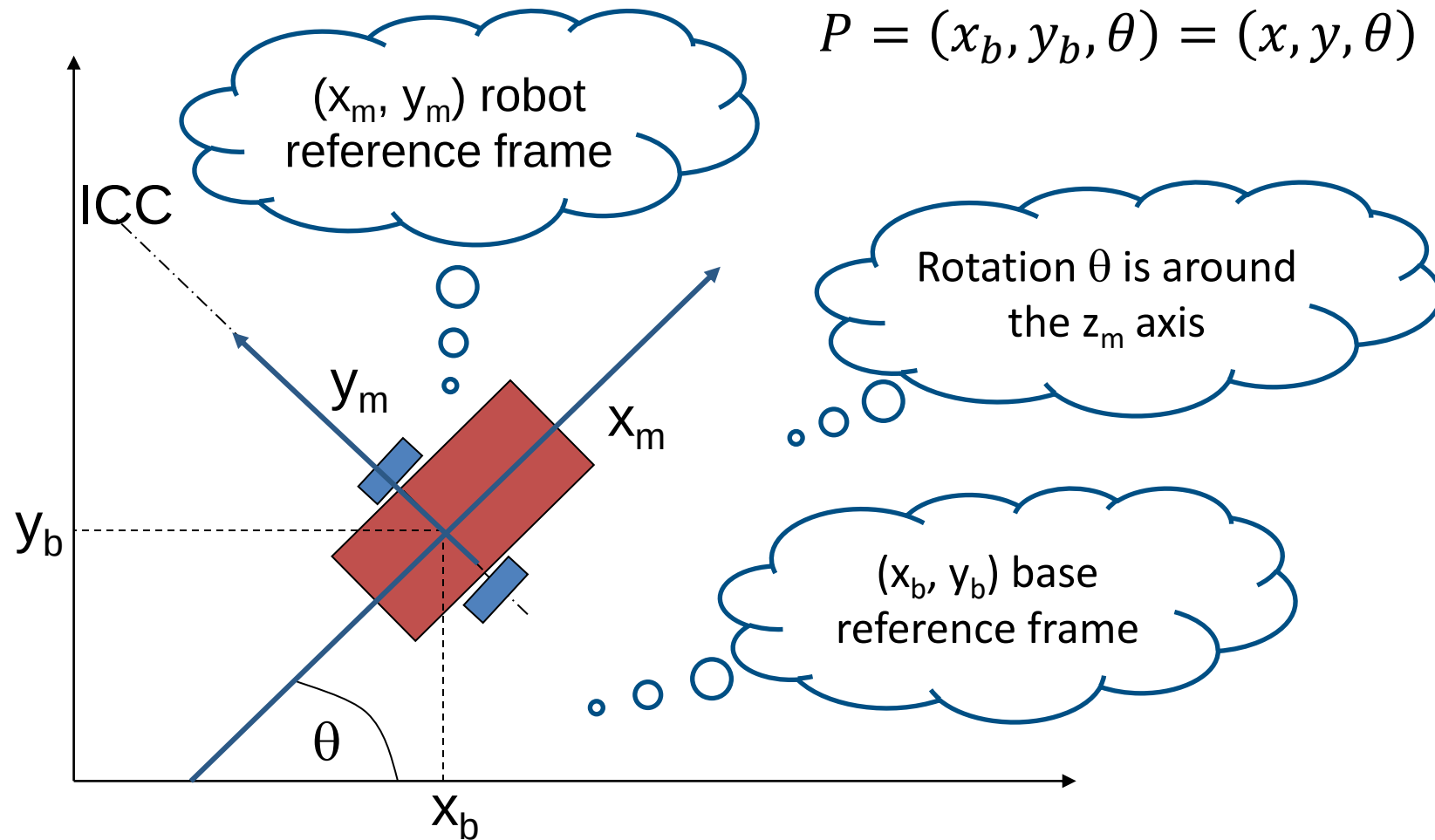
Wheel parameters:
 r = radius
 v = linear velocity
 ω = angular velocity

Instantaneous Center of Curvature (or Rotation)

For a robot to move on the plane (3DoF), without slippage, wheels axis have to intersect in a single point named Instantaneous Center of Curvature (ICC) or Instantaneous Center of Rotation (ICR)



Representing a Pose



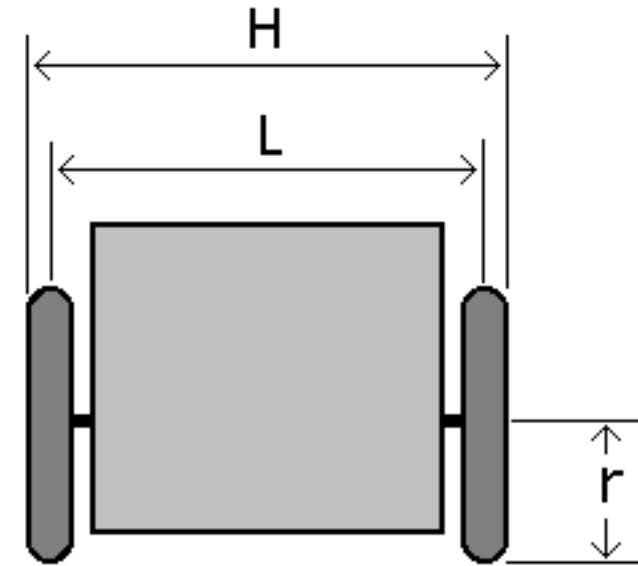
Differential Drive Kinematics (1)

Construction

- 2 wheels on the same axis
- 2 independent motors (one for wheel)
- 3rd passive supporting wheel

Variables independently controlled

- V_R : velocity of the right wheel
- V_L : velocity of the left wheel



Pose representation in base reference:

$$P = (x_b, y_b, \theta) = (x, y, \theta)$$

Control input are:

- v : linear velocity of the robot
- ω : angular velocity of the robot

Linearly related to V_R and V_L ...

Differential Drive Kinematics (2)

Right and left wheels follow a circular path with ω angular velocity and different curvature radius

$$\omega (R + L/2) = V_R$$

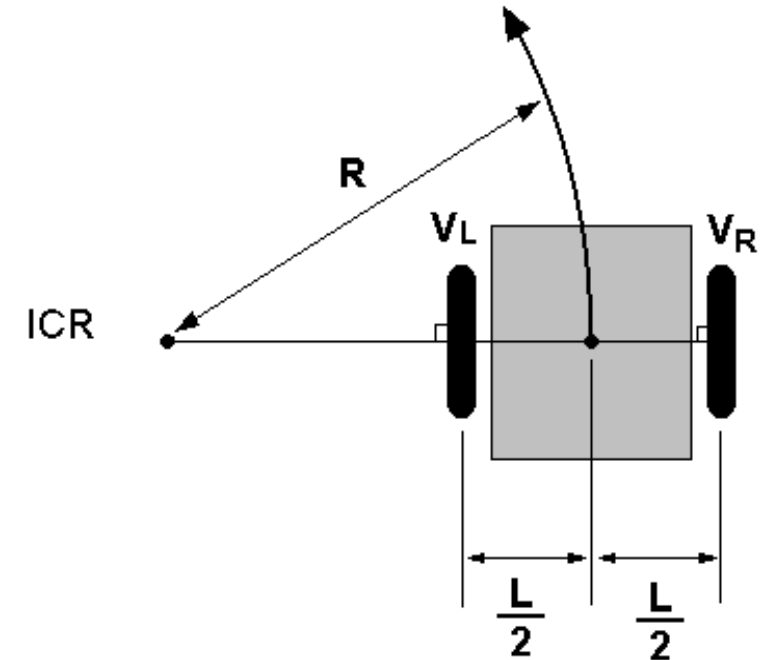
$$\omega (R - L/2) = V_L$$

Given V_R and V_L you can find ω solving for R and equating

$$\omega = (V_R - V_L) / L$$

Similarly you can find R solving for ω and equating

$$R = L/2 (V_R + V_L) / (V_R - V_L)$$

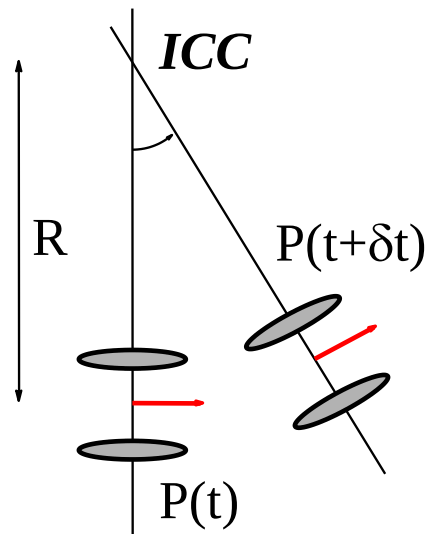
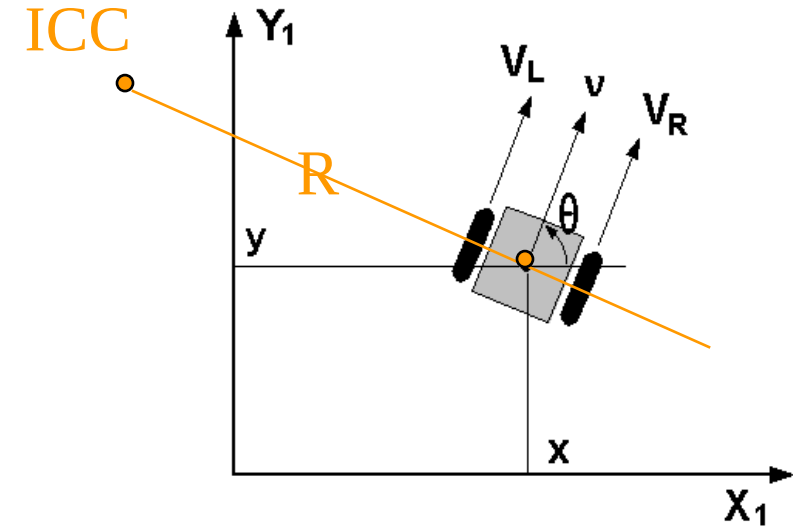


Rotation in place
 $R = 0, V_R = -V_L$
Linear movement
 $R = \text{infinite}, V_R = V_L$

Differential Drive ICC

Wheels move around ICC on a circumference with instantaneous radius R and angular velocity ω

$$ICC = (x + R \cos(\theta + \pi/2), y + R \sin(\theta + \pi/2)) = (x - R \sin(\theta), y + R \cos(\theta))$$



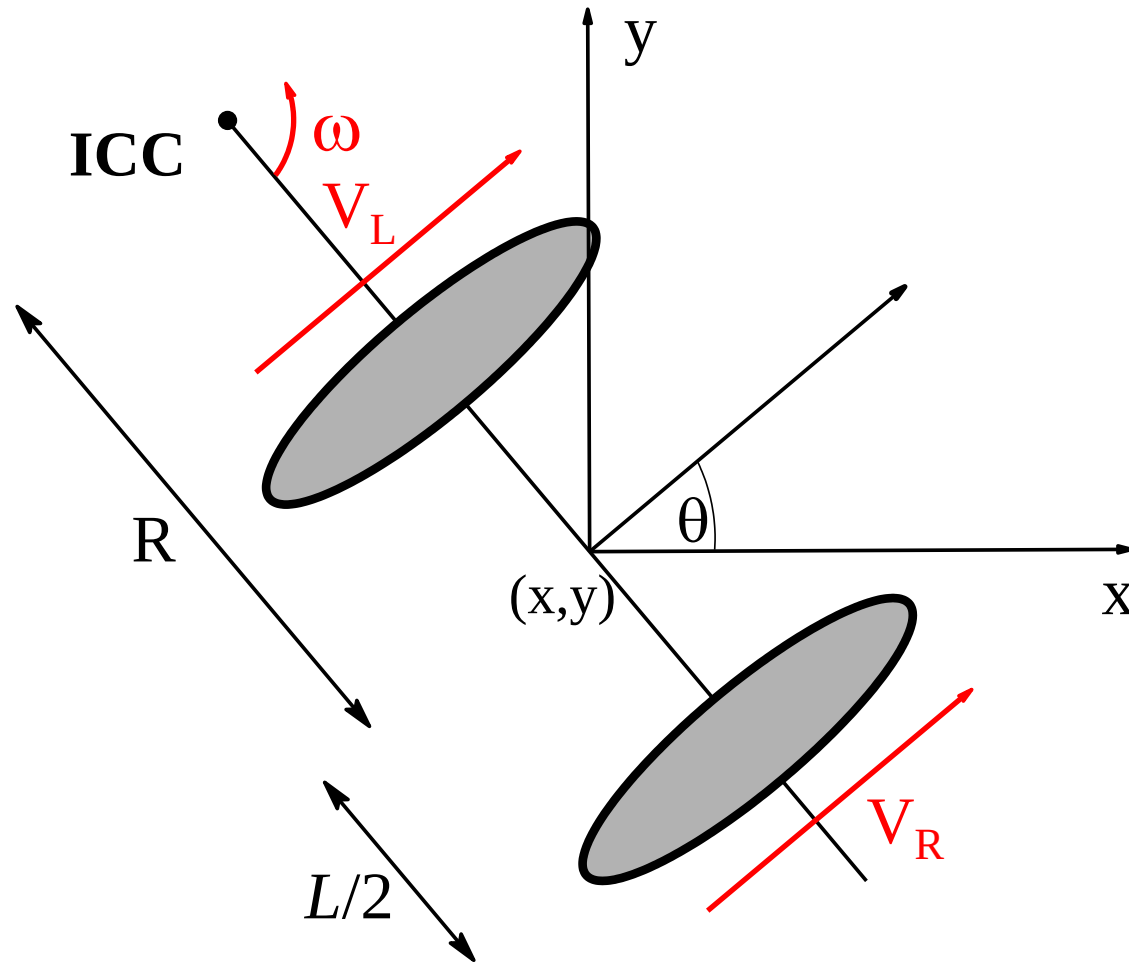
Rotate around ICC

Translate robot in ICC

Translate robot back

$$\begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix} = \begin{bmatrix} \cos(\omega \cdot \delta t) & -\sin(\omega \cdot \delta t) & 0 \\ \sin(\omega \cdot \delta t) & \cos(\omega \cdot \delta t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x - ICC_x \\ y - ICC_y \\ \theta \end{bmatrix} + \begin{bmatrix} ICC_x \\ ICC_y \\ \omega \cdot \delta t \end{bmatrix}$$

Differential Drive Equations (Remember!)



$$ICC = (x - R \cdot \sin(\theta), y + R \cdot \cos(\theta))$$

$$V_R = \omega \cdot (R + L / 2)$$

$$V_L = \omega \cdot (R - L / 2)$$

$$R = \frac{L (V_R + V_L)}{2 (V_R - V_L)}$$

$$V = \frac{V_R + V_L}{2}$$

$$\omega = \frac{V_R - V_L}{L}$$

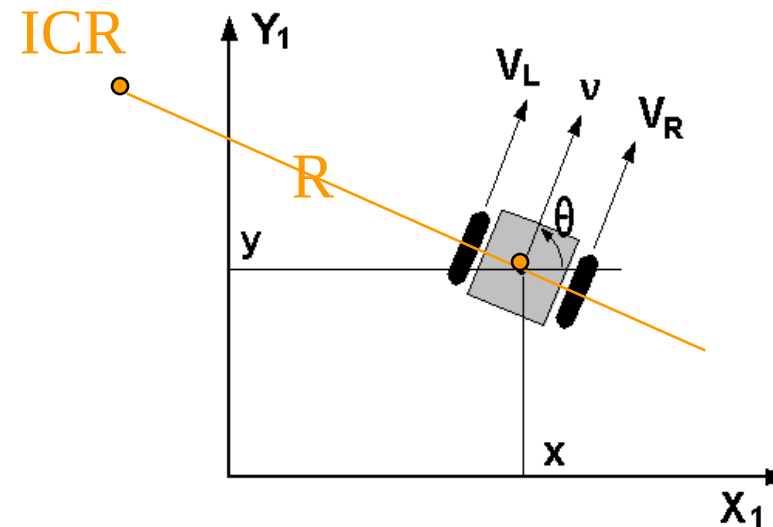
Differential drive odometry

Being known

$$\omega = (V_R - V_L) / L$$

$$R = L/2 (V_R + V_L) / (V_R - V_L)$$

$$V = \omega R = (V_R + V_L) / 2$$



Compute the velocity in the base frame

$$V_x = V(t) \cos(\theta(t))$$

$$V_y = V(t) \sin(\theta(t))$$

Integrate position in base frame

$$x(t) = \int V(t) \cos(\theta(t)) dt$$

$$y(t) = \int V(t) \sin(\theta(t)) dt$$

$$\theta(t) = \int \omega(t) dt$$

Can integrate
at discrete time

$$x(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \cos(\theta(t')) \cdot dt'$$

$$y(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \sin(\theta(t')) \cdot dt'$$

$$\theta(t) = \frac{1}{L} \int_0^t (V_R(t') - V_L(t')) \cdot dt'$$

Odometry Integration (1)

Assume constant linear velocity v_k and angular velocity ω_k in $[t_k, t_{k+1}]$
can use Euler integration to compute robot odometry

$$x(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \cos(\theta(t')) \cdot dt'$$

$$y(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \sin(\theta(t')) \cdot dt'$$

$$\theta(t) = \frac{1}{L} \int_0^t (V_R(t') - V_L(t')) \cdot dt'$$



$$x_{k+1} = x_k + v_k T_S \cos \theta_k$$

$$y_{k+1} = y_k + v_k T_S \sin \theta_k$$

$$\theta_{k+1} = \theta_k + \omega_k T_S$$

$$T_S = t_{k+1} - t_k$$

Approximated

Exact

Odometry Integration (2)

Assume constant linear velocity v_k and angular velocity ω_k in $[t_k, t_{k+1}]$
can use 2nd order Runge-Kutta integration to compute robot odometry

Better
approximation

$$x(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \cos(\theta(t')) \cdot dt'$$

$$y(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \sin(\theta(t')) \cdot dt'$$

$$\theta(t) = \frac{1}{L} \int_0^t (V_R(t') - V_L(t')) \cdot dt'$$



$$x_{k+1} = x_k + v_k T_S \cos\left(\theta_k + \frac{\omega_k T_S}{2}\right)$$

$$y_{k+1} = y_k + v_k T_S \sin\left(\theta_k + \frac{\omega_k T_S}{2}\right)$$

$$\theta_{k+1} = \theta_k + \omega_k T_S$$

$$T_S = t_{k+1} - t_k$$

Average
orientation

Odometry Integration (3)

Assume constant linear velocity v_k and angular velocity ω_k in $[t_k, t_{k+1}]$
can use exact integration to compute the robot odometry

$$x(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \cos(\theta(t')) \cdot dt'$$

$$y(t) = \frac{1}{2} \int_0^t (V_R(t') + V_L(t')) \cdot \sin(\theta(t')) \cdot dt'$$

$$\theta(t) = \frac{1}{L} \int_0^t (V_R(t') - V_L(t')) \cdot dt'$$



$$x_{k+1} = x_k + \frac{v_k}{\omega_k} (\sin \theta_{k+1} - \sin \theta_k)$$

$$y_{k+1} = y_k - \frac{v_k}{\omega_k} (\cos \theta_{k+1} - \cos \theta_k)$$

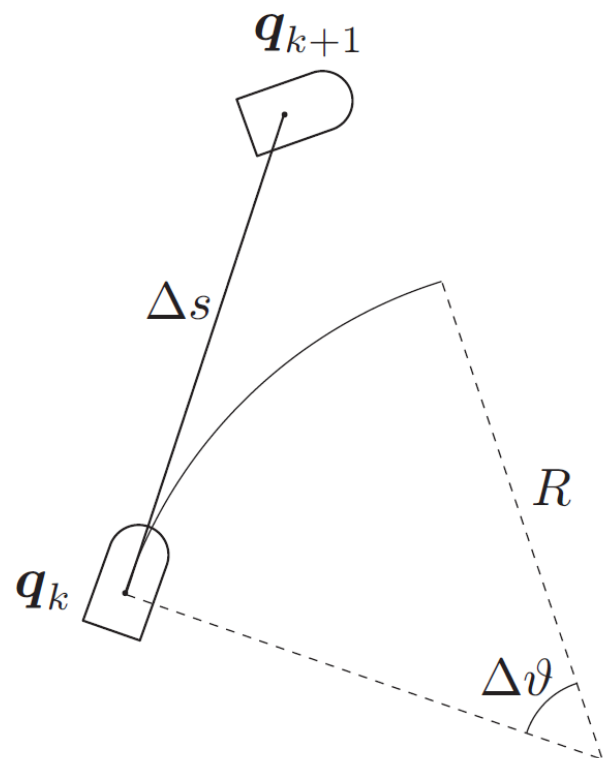
$$\theta_{k+1} = \theta_k + \omega_k T_S$$

$$T_S = t_{k+1} - t_k$$

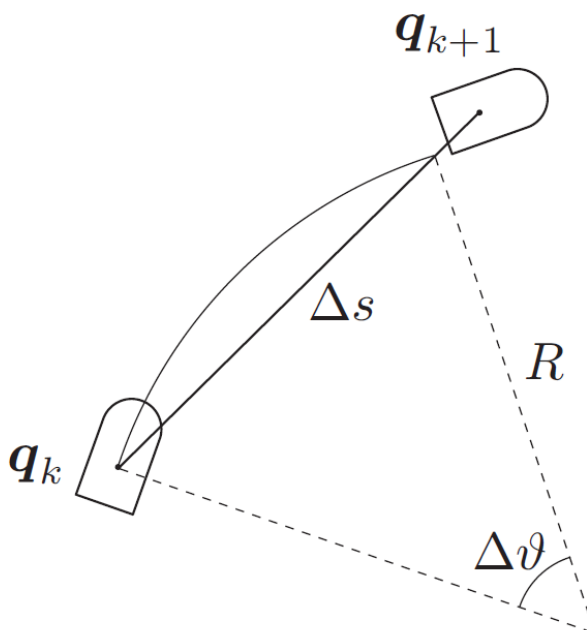
Exact

Need to use Runge
Kutta for $\omega \sim 0$

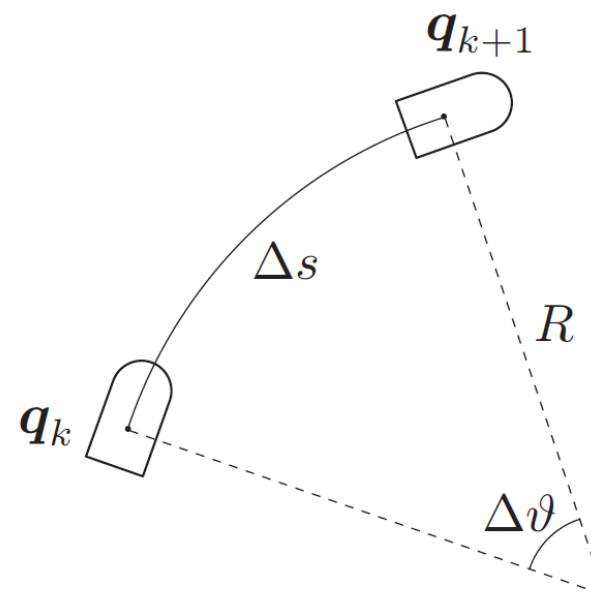
Odometry Comparison



Euler



Runge-Kutta



exact

Tips and Tricks

Proprioceptive measurements are used to compute linear v_k and angular velocity ω_k

$$v_k T_S = \Delta s, \quad \omega_k T_S = \Delta \theta, \quad \frac{v_k}{\omega_k} = \frac{\Delta s}{\Delta \theta}$$

with Δs is the traveled distance and $\Delta \theta$ is the orientation change

In a differential drive these become

$$\Delta s = \frac{r}{2} (\Delta \phi_R + \Delta \phi_L), \quad \Delta \theta = \frac{r}{L} (\Delta \phi_R - \Delta \phi_L)$$

with $\Delta \phi_R$ and $\Delta \phi_L$ the total rotations measured by wheel encoders

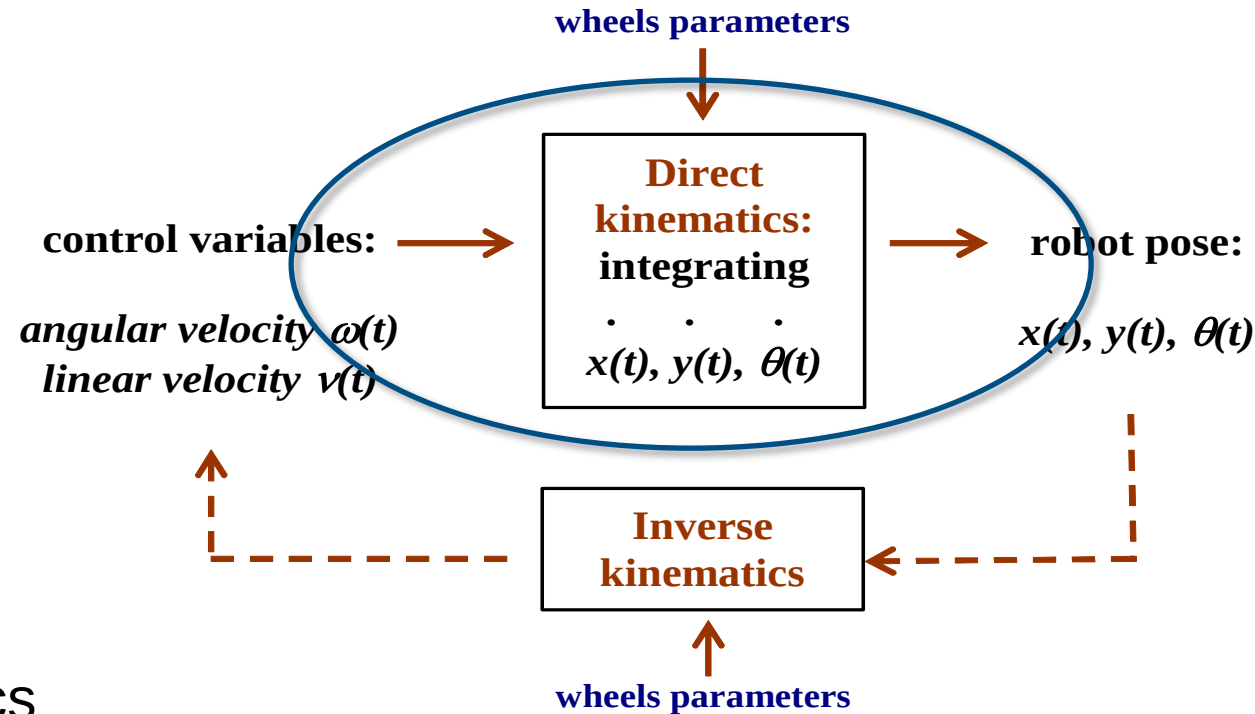
Nice but drifts
because of ...

... slippage ...

... integration...

... calibration ...





Direct kinematics

- Given control parameters, e.g., wheels and velocities, and a time of movement t , find the pose (x, y, θ) reached by the robot

Inverse kinematics

- Given the final pose (x, y, θ) find control parameters to move there in a given time t

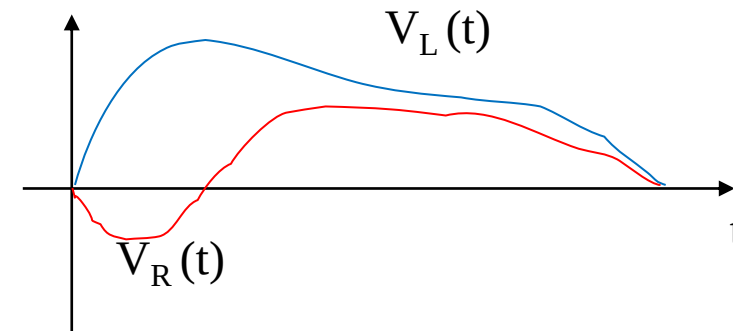
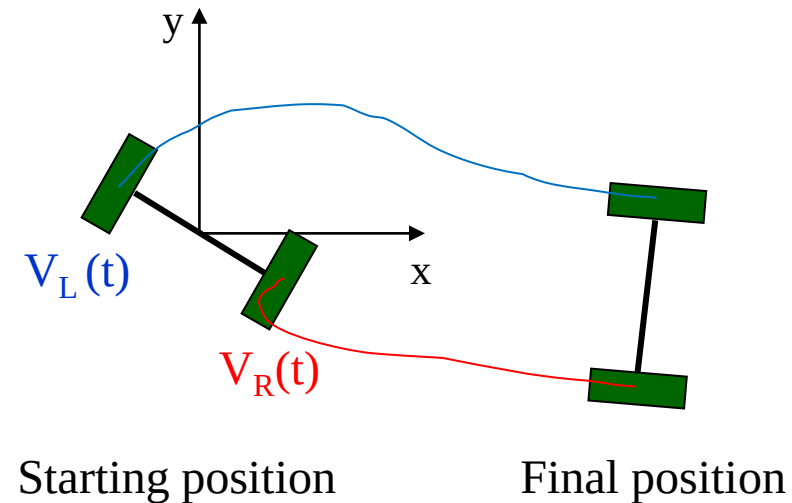
Inverse kinematics

Given a desired position or velocity, what can we do to achieve it?

Finding “some” solution is not hard, but finding the “best” solution can be very difficult:

- Shortest time
- Most energy efficient
- Smoothest velocity profiles

Moreover if we have non holonomic constraints and only two control variables; we cannot directly reach any of the 3DoF final positions ...



Differential drive inverse kinematics

Decompose the problem and control only few DoF at the time

1. Turn so that the wheels are parallel to the line between the original and final position of robot origin

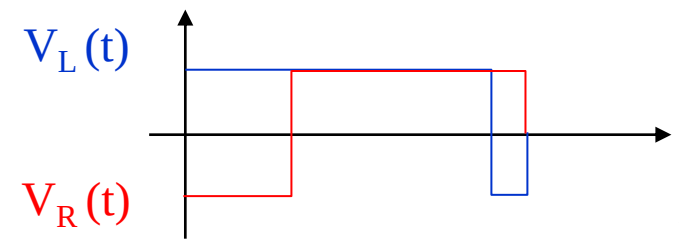
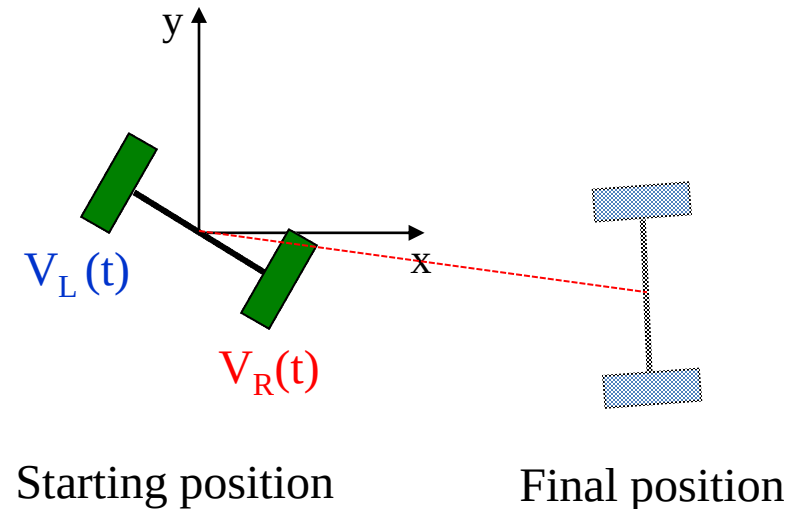
$$-V_L(t) = V_R(t) = V_{\max}$$

2. Drive straight until the robot's origin coincides with destination

$$V_L(t) = V_R(t) = V_{\max}$$

3. Rotate again in to achieve the desired final orientation

$$-V_L(t) = V_R(t) = V_{\max}$$



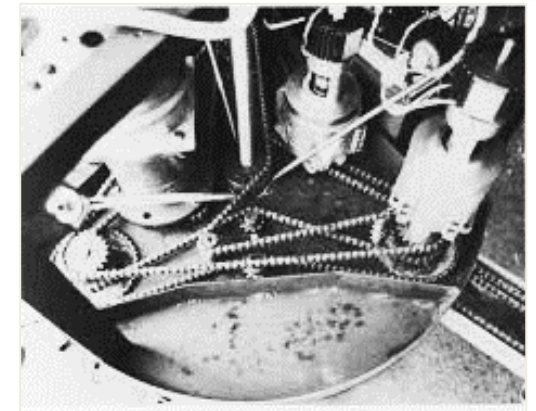
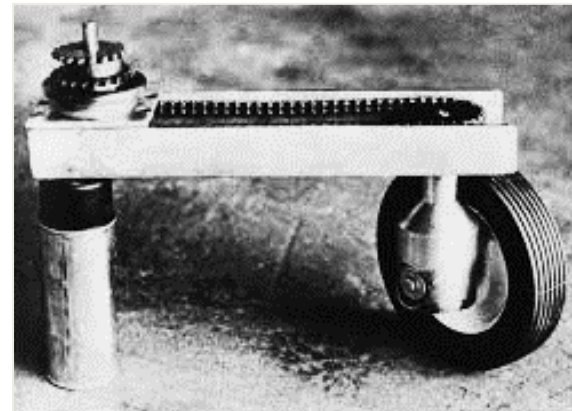
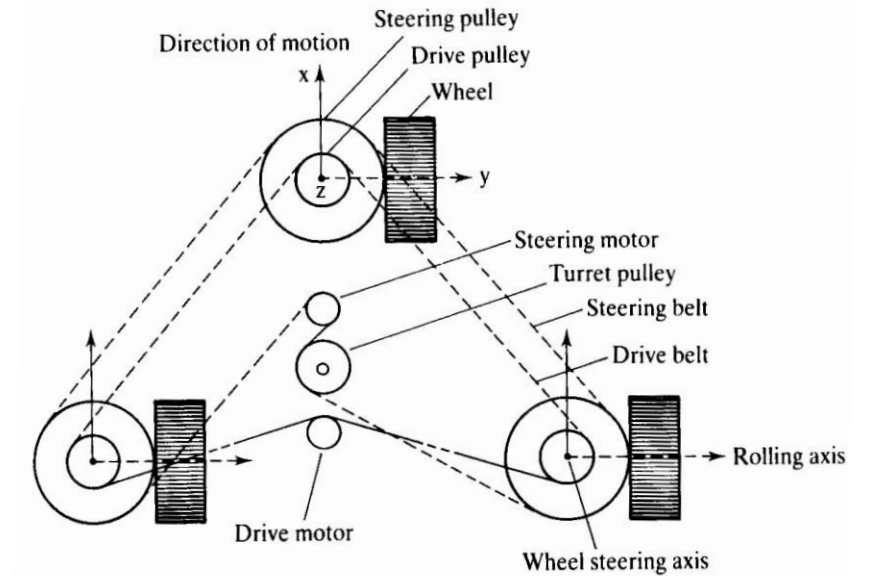
Synchronous drive

Complex mechanical robot design

- (At least) 3 wheels actuated and steered
- A motor to roll all the wheels, a second motor to rotate them
- Wheels point in the same direction
- It is possible to control directly θ with an additional actuator

Robot control variables

- Linear velocity $v(t)$
- Angular velocity $\omega(t)$



Its ICC is always at the infinite and the robot is non holonomic (it can only translate freely)

Synchronous drive kinematics

Robot control for the synchronous drive

- Direct control of $v(t)$ and $\omega(t)$
- Steering changes the direction of ICC

Particular cases:

- $v(t)=0$, $\omega(t) = \omega$ for $dt \rightarrow$ robot rotates in place
- $v(t)=v$, $\omega(t) = 0$ for $dt \rightarrow$ robot moves linearly

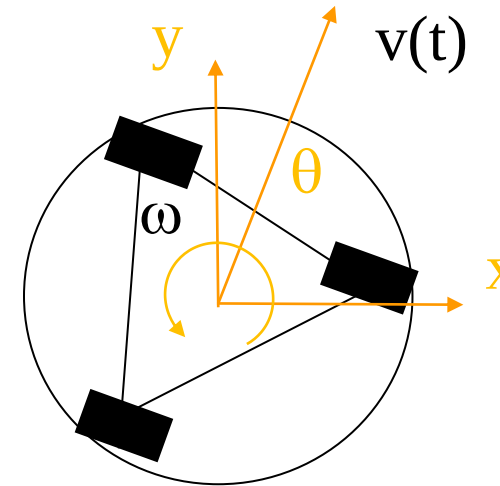
Compute the velocity in the base frame

$$V_x = V(t) \cos(\theta(t))$$

$$V_y = V(t) \sin(\theta(t))$$

Calls odometry
also for diff drive!

Integrate position in base frame to get the robot odometry (traversed path) ...



$$\begin{aligned} x(t) &= \int_0^t v(t') \cos[\theta(t')] dt' \\ y(t) &= \int_0^t v(t') \sin[\theta(t')] dt' \\ \theta(t) &= \int_0^t \omega(t') dt' \end{aligned}$$

Synchro drive inverse kinematics

Decompose the problem and control only a few degrees of freedom at a time

1. Turn so that the wheels are parallel to the line between the original and final position of robot origin

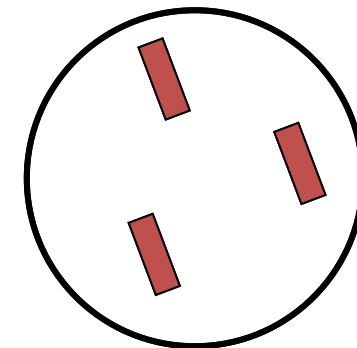
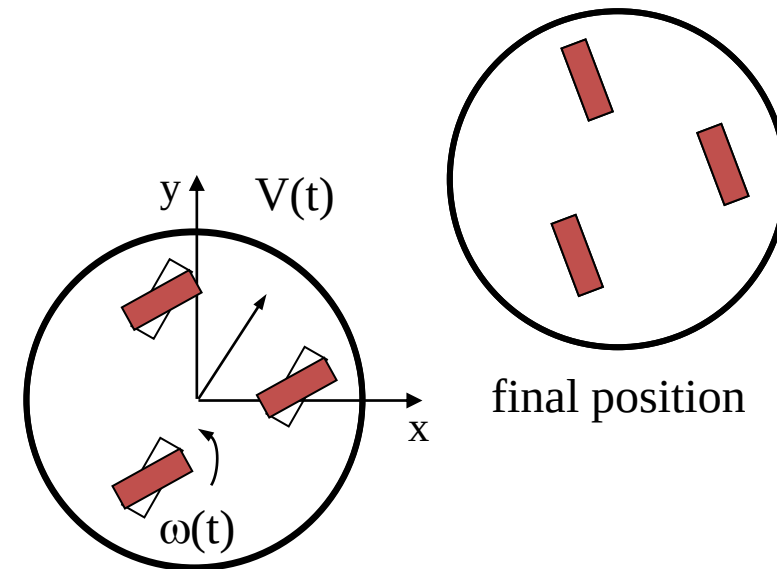
$$\omega(t) = \omega_{\max}$$

2. Drive straight until the robot's origin coincides with destination

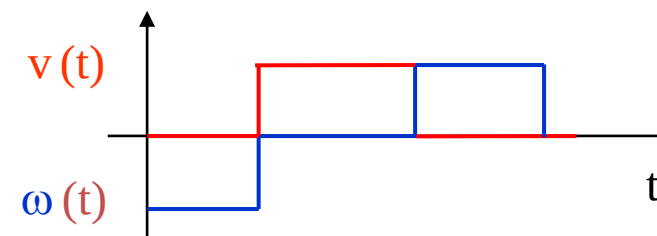
$$v(t) = v_{\max}$$

3. Rotate again in to achieve the desired final orientation

$$\omega(t) = \omega_{\max}$$



starting position



Omnidirectional (Syncro drive)



Omnidirectional Robot

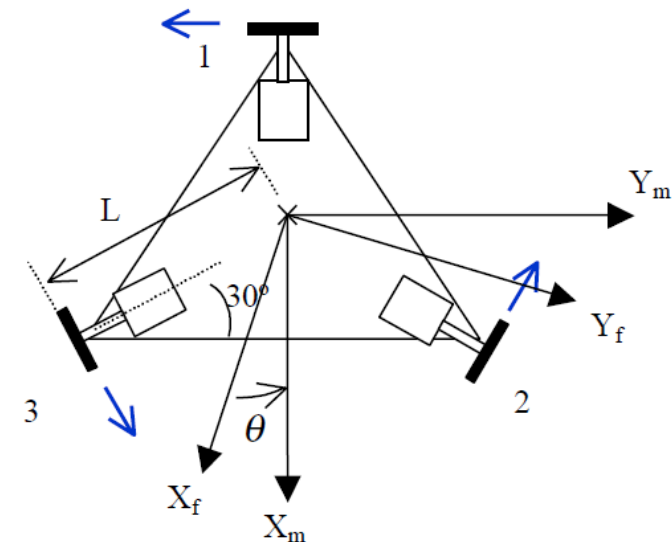
Simple mechanical robot design

- (At least) 3 Swedish wheels actuated
- One independent motor per wheel
- Wheels point in different direction
- It is possible to control directly x, y, θ

Robot control variables

- Linear velocity $v(t)$ (each component)
- Angular velocity $\omega(t)$

$$\begin{bmatrix} V_x \\ V_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{\sqrt{3}}r & \frac{1}{\sqrt{3}}r \\ -\frac{2}{3}r & \frac{1}{3}r & \frac{1}{3}r \\ \frac{r}{3L} & \frac{r}{3L} & \frac{r}{3L} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$



Omnidirectional (Swedish wheels)



Tricycle kinematics

The Tricycle is the typical kinematics of AGV

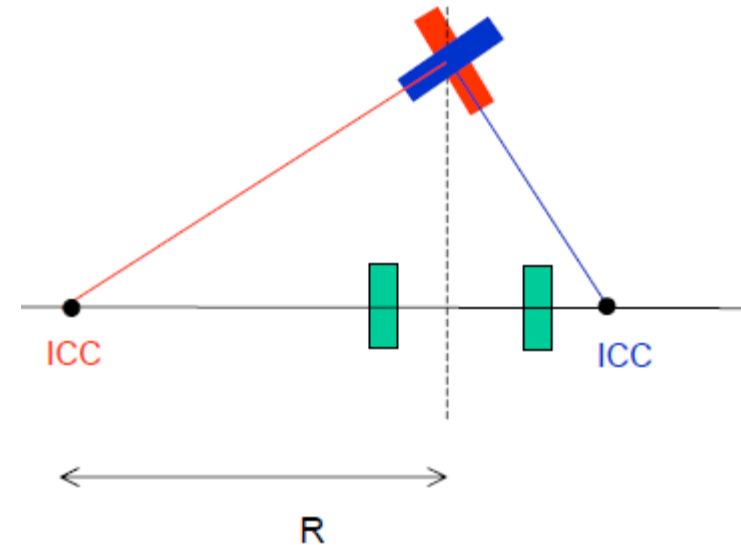
- One actuated and steerable wheel
- 2 additional passive wheels
- Cannot control θ independently unless $\alpha(t)$ can be up to 90 degrees
- ICC must lie on the line that passes through the fixed wheels

Robot control variables

- Steering direction $\alpha(t)$
- Angular velocity of steering wheel $\omega_s(t)$

Particular cases:

- $\alpha(t)=0$, $\omega_s(t) = \omega \rightarrow$ moves straight
- $\alpha(t)=90$, $\omega_s(t) = \omega \rightarrow$ rotates in place



Tricycle kinematics

Direct kinematics can be derived as:

$r = \text{steering wheel radius}$

$$V_s(t) = \omega_s(t) \cdot r$$

$$R(t) = d \cdot \tan\left(\frac{\pi}{2} - \alpha(t)\right)$$

$$\omega(t) = \frac{\omega_s(t) \cdot r}{\sqrt{d^2 + R(t)^2}} = \frac{V_s(t)}{d} \sin \alpha(t)$$

Angular velocity of the moving frame

Linear velocity $v(t)$

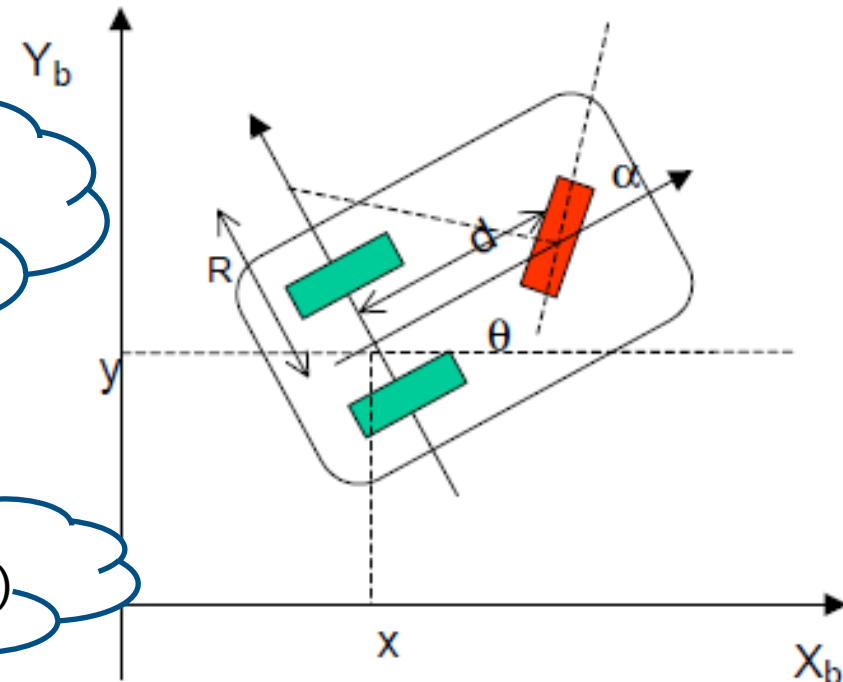
In the robot frame

$$V_x(t) = V_s(t) \cdot \cos \alpha(t)$$

We assume no slippage

$$\dot{\theta} = \frac{V_s(t)}{d} \cdot \sin \alpha(t)$$

Angular velocity $\omega(t)$



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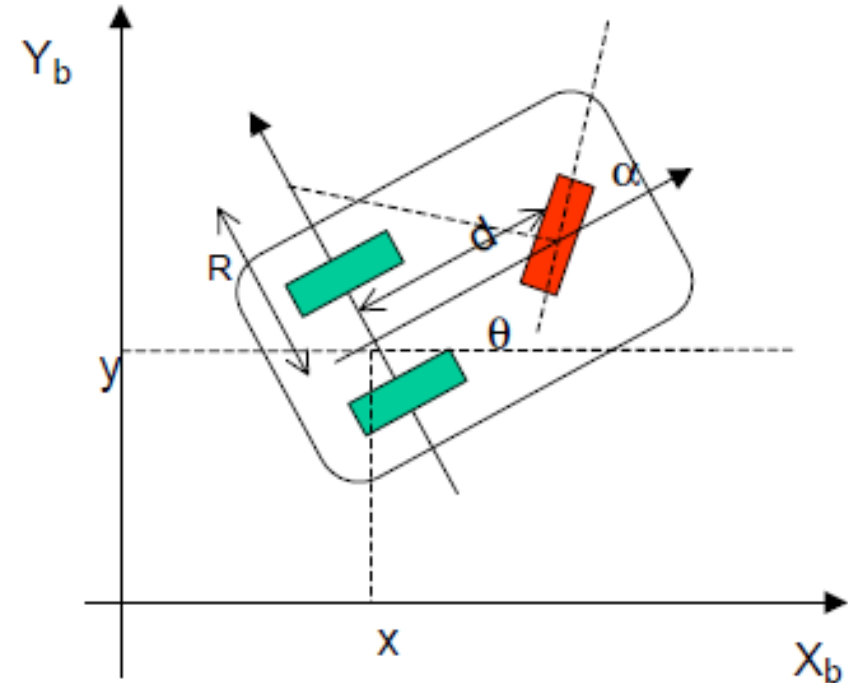
$$\omega(t) = \frac{\omega_s(t) \cdot r}{\sqrt{d^2 + R(t)^2}} = \frac{V_s(t)}{d} \sin \alpha(t)$$

In the base frame

$$\dot{x}(t) = V_s(t) \cdot \cos \alpha(t) \cdot \cos \theta(t) = V(t) \cdot \cos \theta(t)$$

$$\dot{y}(t) = V_s(t) \cdot \cos \alpha(t) \cdot \sin \theta(t) = V(t) \cdot \sin \theta(t)$$

$$\dot{\theta} = \frac{V_s(t)}{d} \cdot \sin \alpha(t) = \omega(t)$$



Ackerman steering

Most diffused kinematics on the planet

- Four wheels turning
- Wheels have limited turning angles
- No in-place rotation

Similar to the Tricycle model

$$R = \frac{d}{\tan \alpha_R} + b$$

$$\frac{\omega d}{\sin \alpha_R} = V_{FR}$$

Determines angular velocity $\omega(t)$

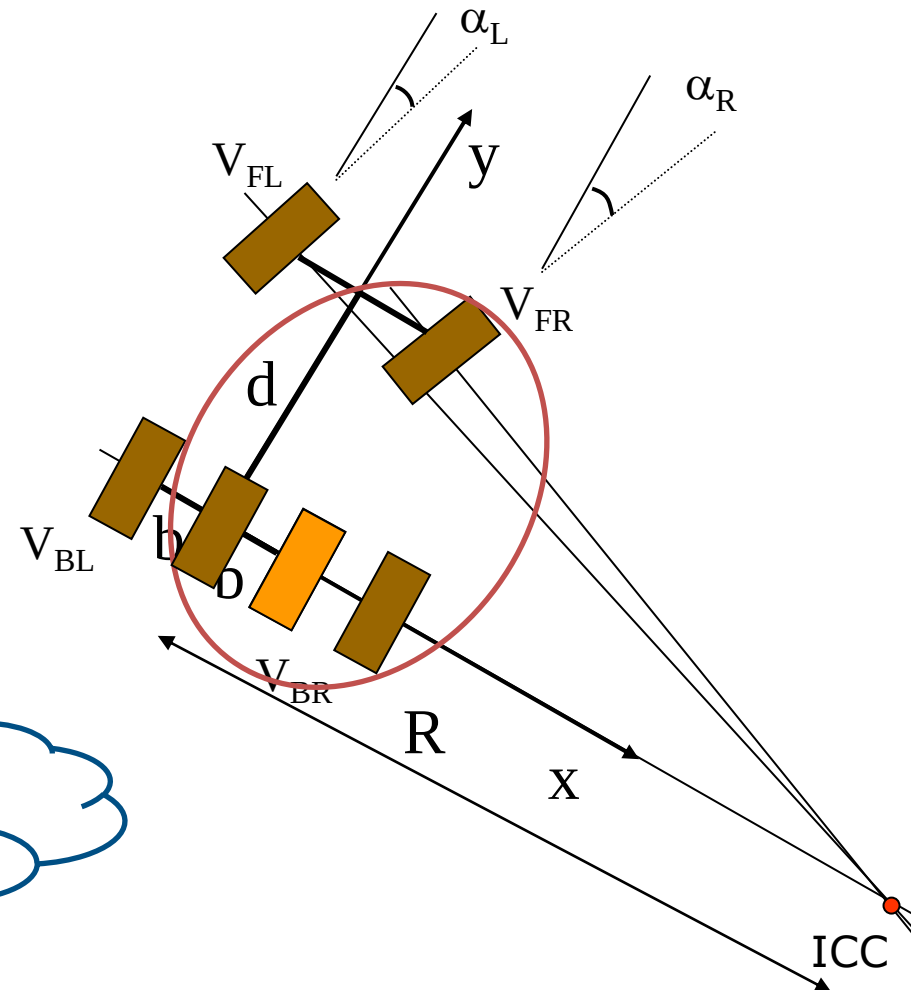
Derive the rest as:

$$\frac{\omega d}{\sin \alpha_L} = V_{FL}$$

$$\alpha_L = \tan^{-1}\left(\frac{d}{R + b}\right)$$

$$\omega(R + b) = V_{BL}$$

$$\omega(R - b) = V_{BR}$$



Ackerman steering (bicycle approximation)

Most diffused kinematics on the planet

- Four wheels turning
- Wheels have limited turning angles
- No in-place rotation

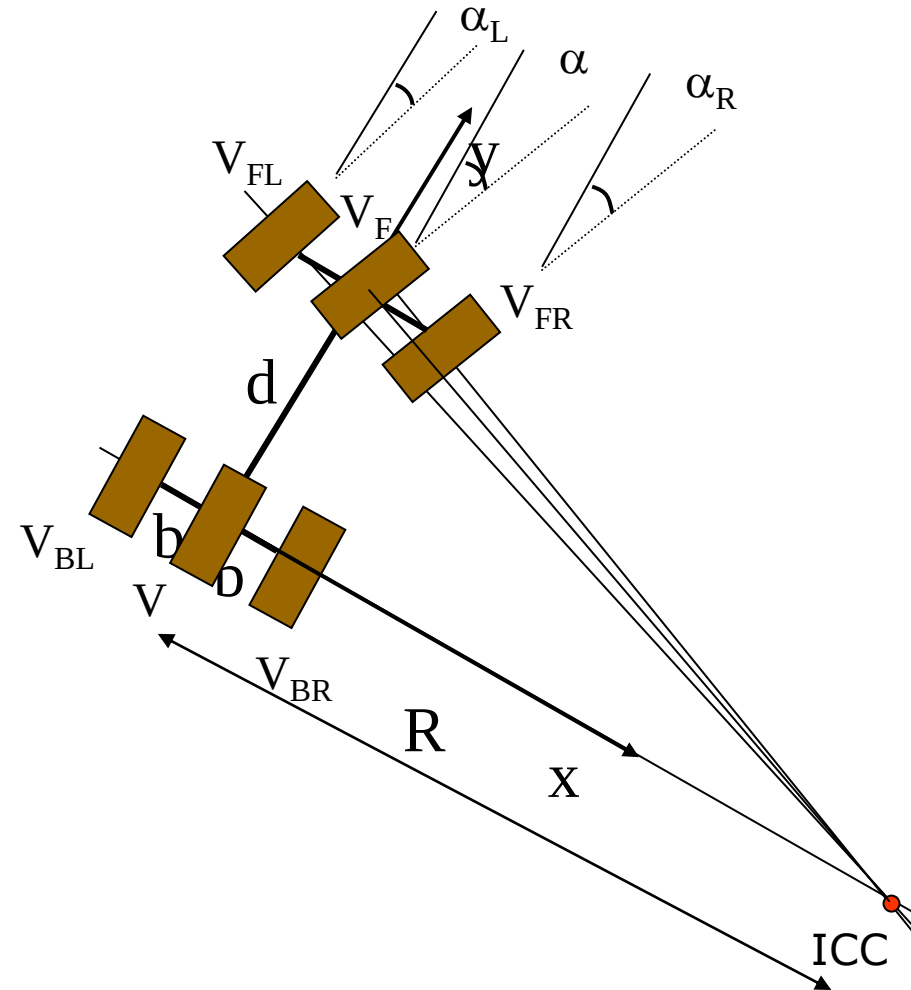
Bicycle approximation

$$R = \frac{d}{\tan \alpha}$$

$$\frac{\omega d}{\sin \alpha} = V_F$$

Referred to the center of real wheels

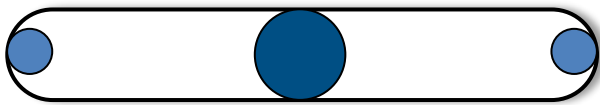
$$\omega R = V \quad \Rightarrow \quad \omega = V \cdot \frac{\tan \alpha}{d}$$



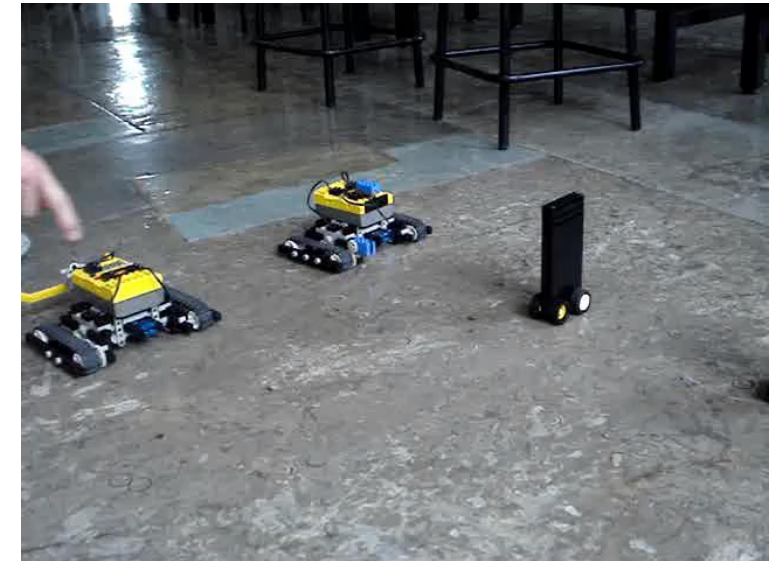
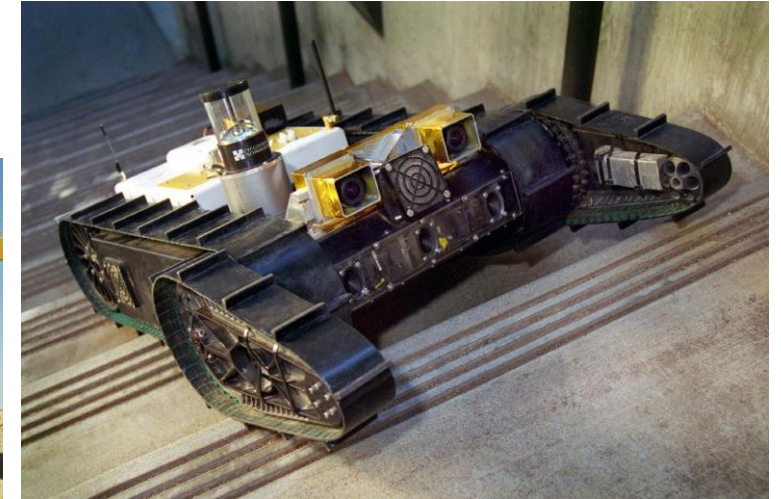
Vehicles with tracks

Vehicles with track have a kinematics similar to the differential drive

- Speed control of each track
- Use the height of the track as wheel diameter



- Often named Skid Steering



Need proper calibration and slippage modeling ...

Skid Steering (approximate) Kinematics

Let' assume:

- Mass in the center of the vehicle
- All wheels in contact with the ground
- Wheels on the same side have same speed

$$\omega_l = \omega_1 = \omega_2 \quad \omega_r = \omega_3 = \omega_4$$

While moving we have multiple ICR and all share ω_z

$$y_G = \frac{v_x}{\omega_z}$$

$$y_l = \frac{v_x - \omega_l r}{\omega_z}$$

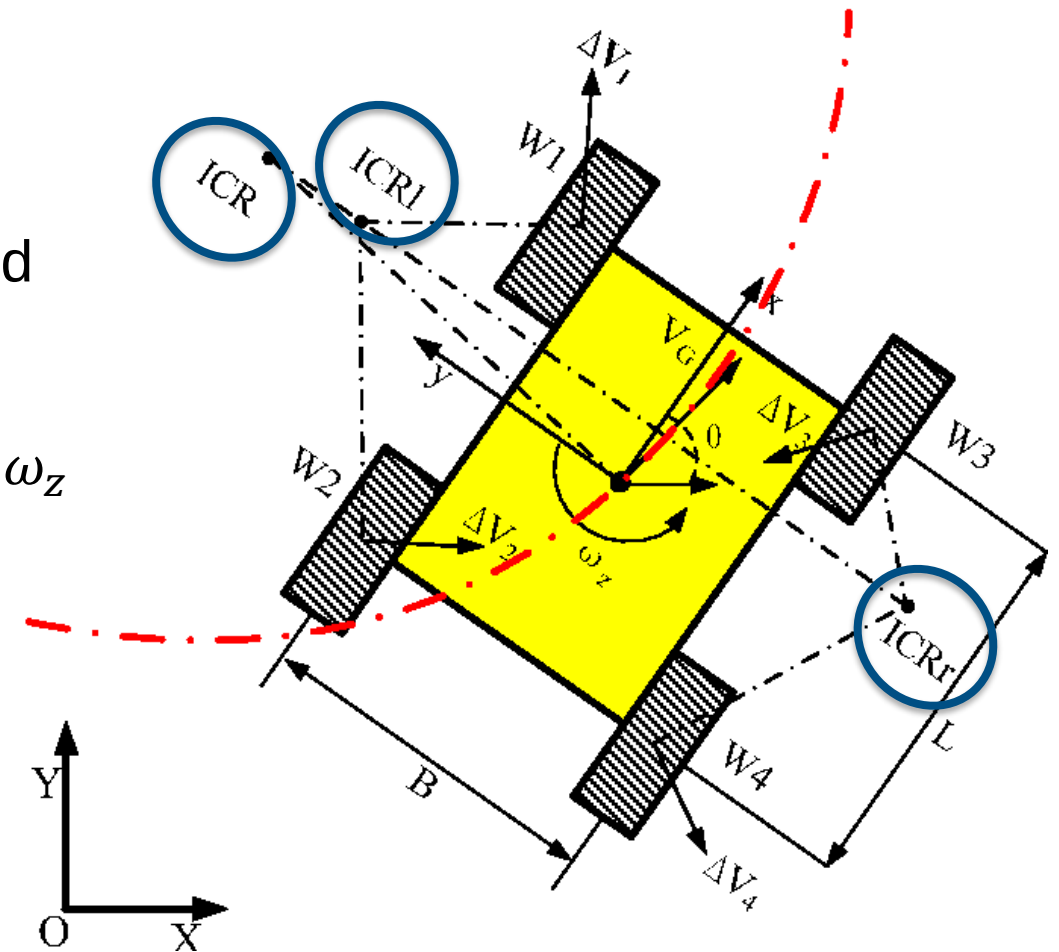
$$y_r = \frac{v_x - \omega_r r}{\omega_z}$$

$$x_G = x_l = x_r = -\frac{v_y}{\omega_z}$$



$$\begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix} = J_\omega \begin{bmatrix} \omega_l r \\ \omega_r r \end{bmatrix}$$

$$J_\omega = \frac{1}{y_l - y_r} \begin{bmatrix} -y_r & y_l \\ x_G & -x_G \\ -1 & 1 \end{bmatrix}$$



Skid Steering (approximate) Kinematics

Let' assume:

- Mass in the center of the vehicle
- All wheels in contact with the ground
- Wheels on the same side have same speed

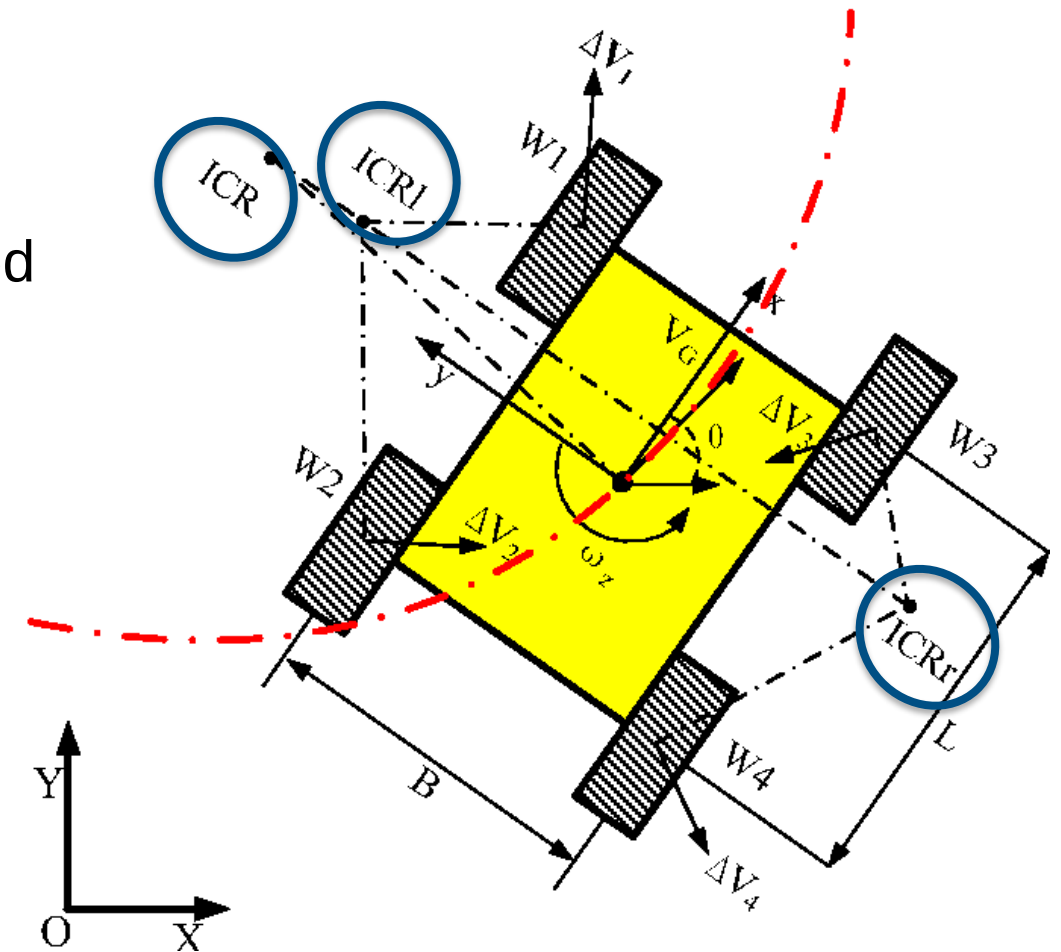
$$\omega_l = \omega_1 = \omega_2 \quad \omega_r = \omega_3 = \omega_4$$

Assume the robot is symmetric

$$\begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix} = J_\omega \begin{bmatrix} \omega_l r \\ \omega_r r \end{bmatrix}$$

$$J_\omega = \frac{1}{2y_0} \begin{bmatrix} y_0 & y_0 \\ x_G & -x_G \\ -1 & 1 \end{bmatrix} \rightarrow \begin{cases} v_x = \frac{v_l + v_r}{2} \\ v_y = 0 \\ \omega_z = \frac{-v_l + v_r}{2y_0} \end{cases}$$

$$y_0 = y_l = -y_r$$



Skid Steering (approximate) Kinematics

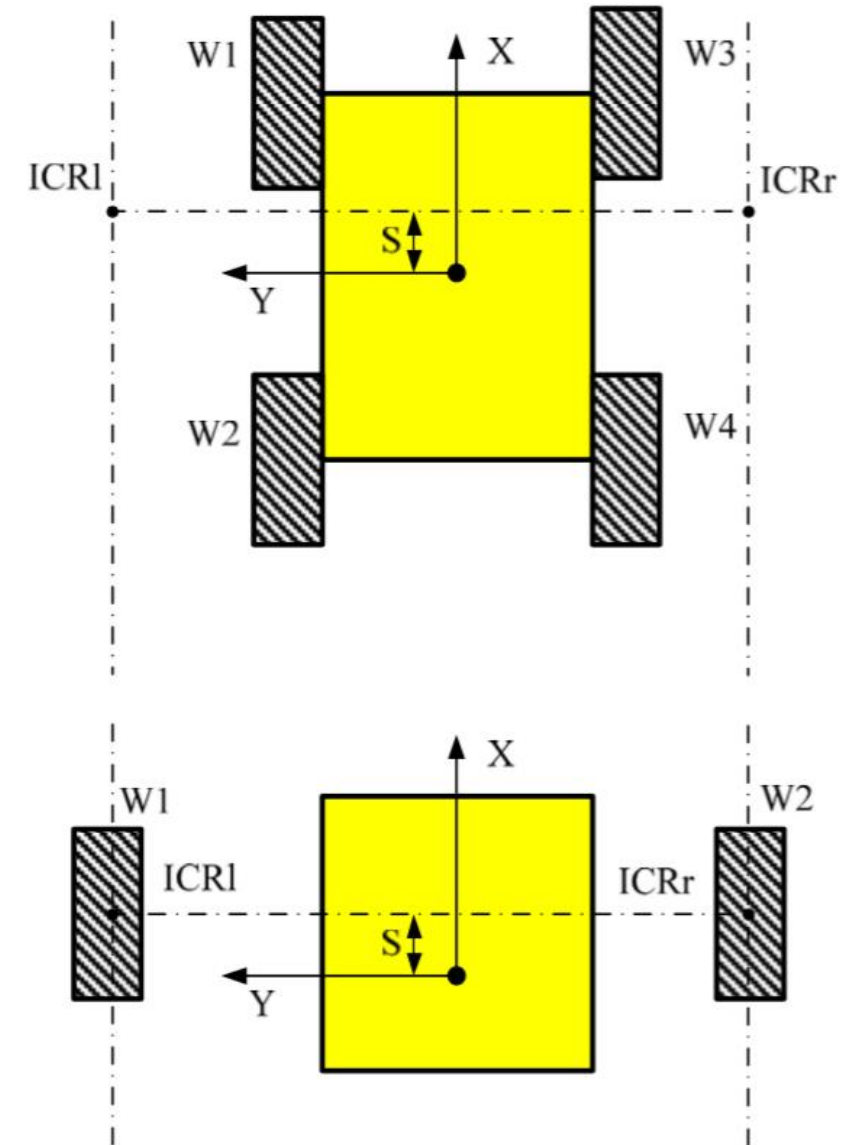
Let' assume:

- Mass in the center of the vehicle
- All wheels in contact with the ground
- Wheels on the same side have same speed

$$\omega_l = \omega_1 = \omega_2 \quad \omega_r = \omega_3 = \omega_4$$

We can get the instantaneous radius of curvature

$$\left\{ \begin{array}{l} v_x = \frac{v_l + v_r}{2} \\ v_y = 0 \\ \omega_z = \frac{-v_l + v_r}{2y_0} \end{array} \right. \Rightarrow \begin{array}{l} R = \frac{v_G}{\omega_z} = \frac{v_l + v_r}{-v_l + v_r} y_0 \\ \lambda = \frac{v_l + v_r}{-v_l + v_r} \\ \chi = \frac{y_l - y_r}{B} = \frac{2y_0}{B}, \quad \chi \geq 1 \end{array}$$



Mobile robots beyond the wheels

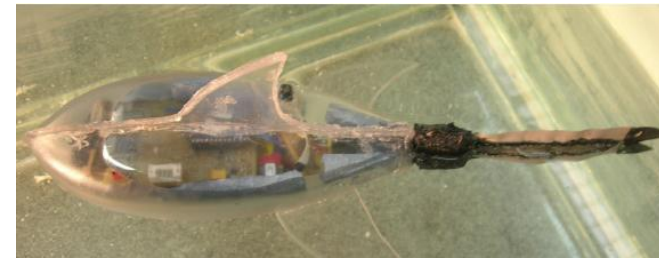
To move on the ground

- Multiple wheels
- Whegs
- Legs



To move in water

- Torpedo-like (single propeller)
- Bodies with thrusters
- Bioinspired



To move in air

- Fixed wings vehicles
- Mobile wings vehicles
- Multi-rotors

